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Lee et al.

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(54) **SEMICONDUCTOR DEVICES AND DATA STORAGE SYSTEMS INCLUDING THE SAME**

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(58) **Field of Classification Search**

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See application file for complete search history.

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Primary Examiner — Alia Sabur

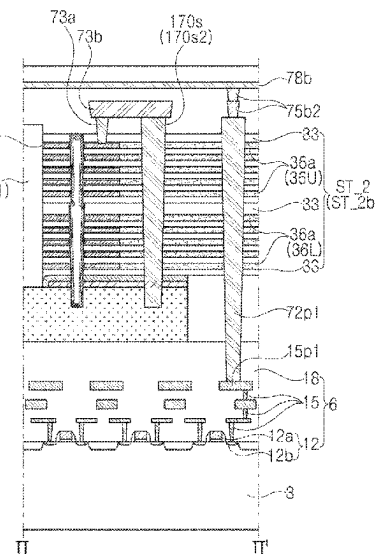
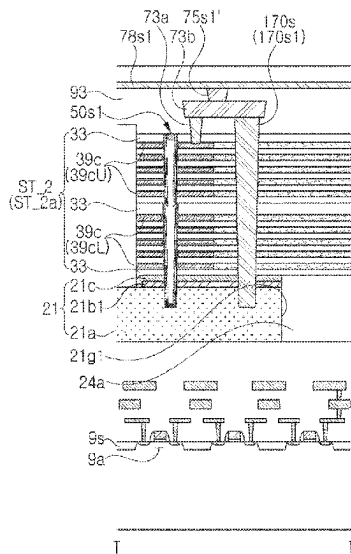
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(57)

ABSTRACT

A semiconductor device may include: a semiconductor substrate; a peripheral circuit structure on the semiconductor substrate; a plate pattern on the peripheral circuit structure and having a gap; and a stack structure on the plate pattern and including a first stack region and a second stack region. The first stack region may include gate electrodes stacked in a vertical direction perpendicular to an upper surface of the semiconductor substrate, and the second stack region may include both a conductor stack region including conductive layers stacked in the vertical direction and an insulator stack region including molded insulating layers at substantially the same height level as the conductive layers. The semiconductor device may also include vertical memory structure that extends through the first stack region; and source contact plugs electrically connected to at least one of the conductive layers of the conductor stack region and contacting the plate pattern.

20 Claims, 27 Drawing Sheets



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<i>H10B 43/35</i>	(2023.01)
<i>H10B 43/50</i>	(2023.01)

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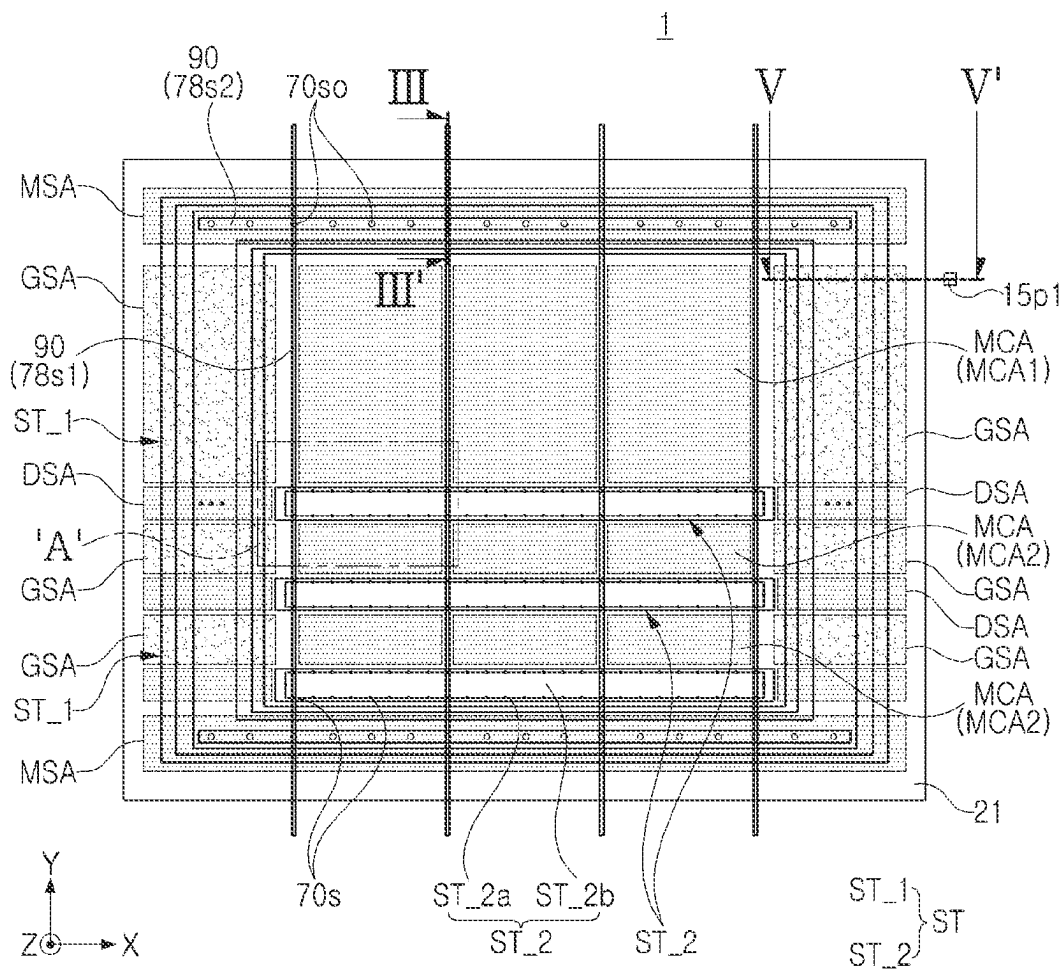
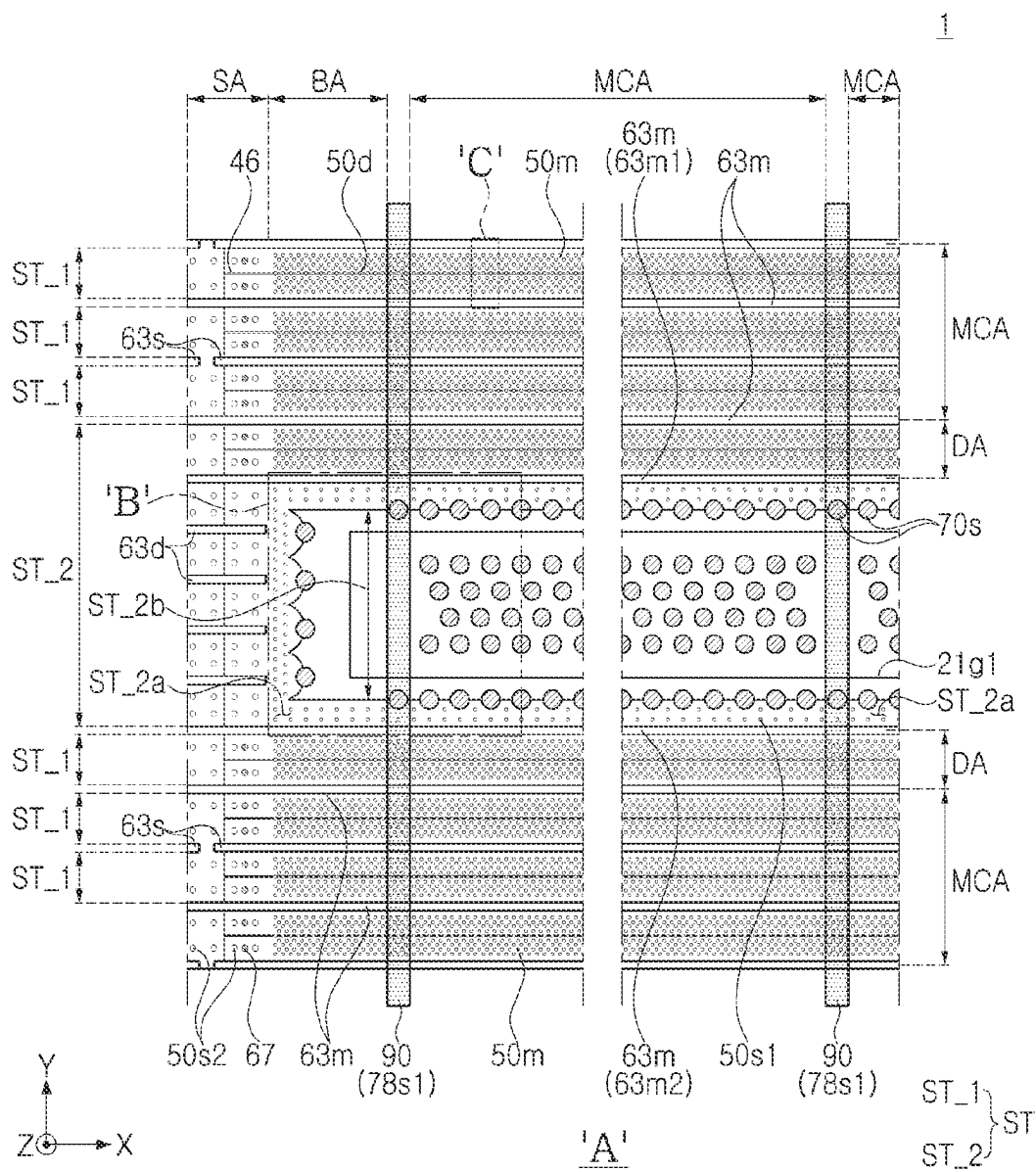


FIG. 1



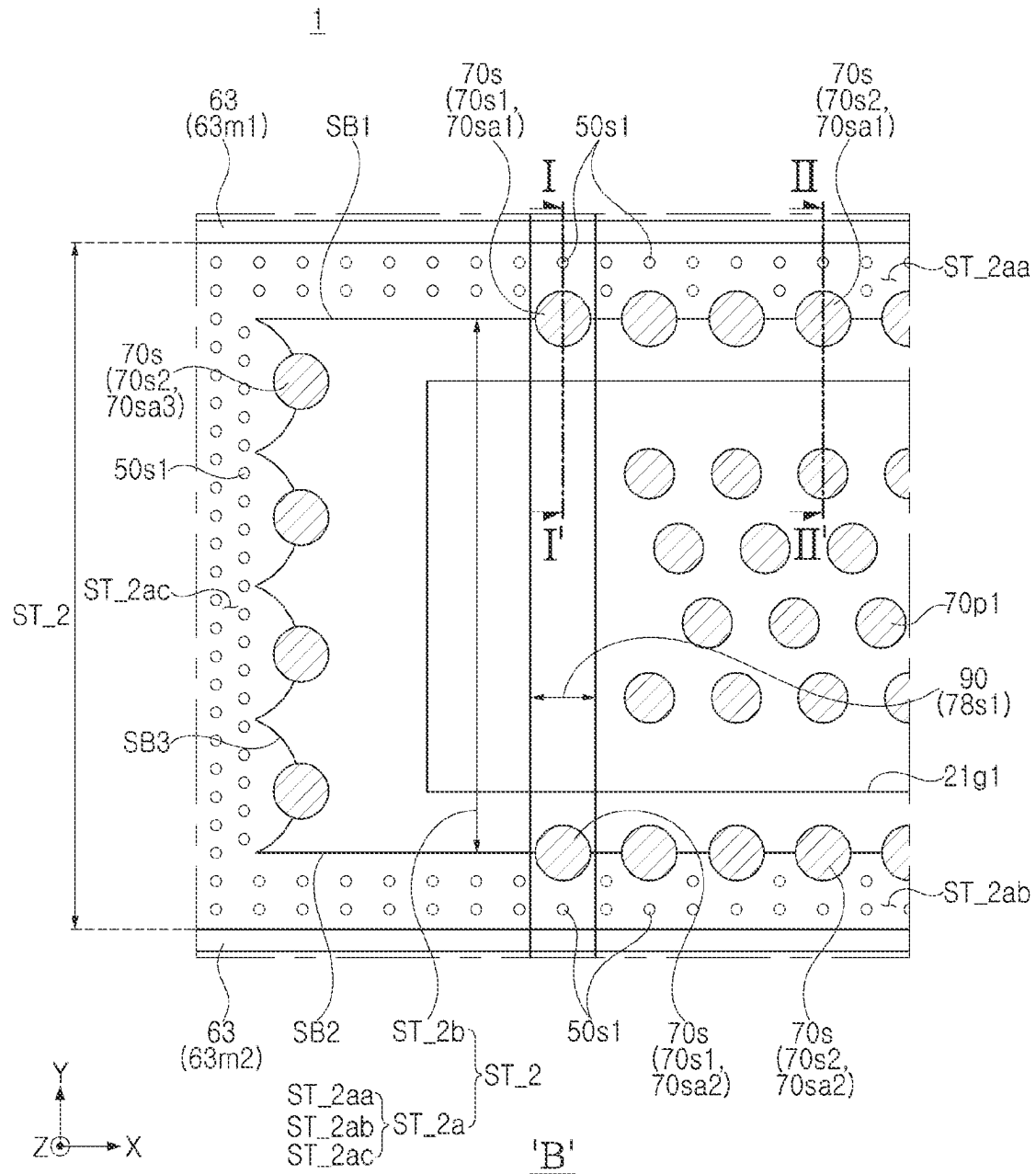


FIG. 3A

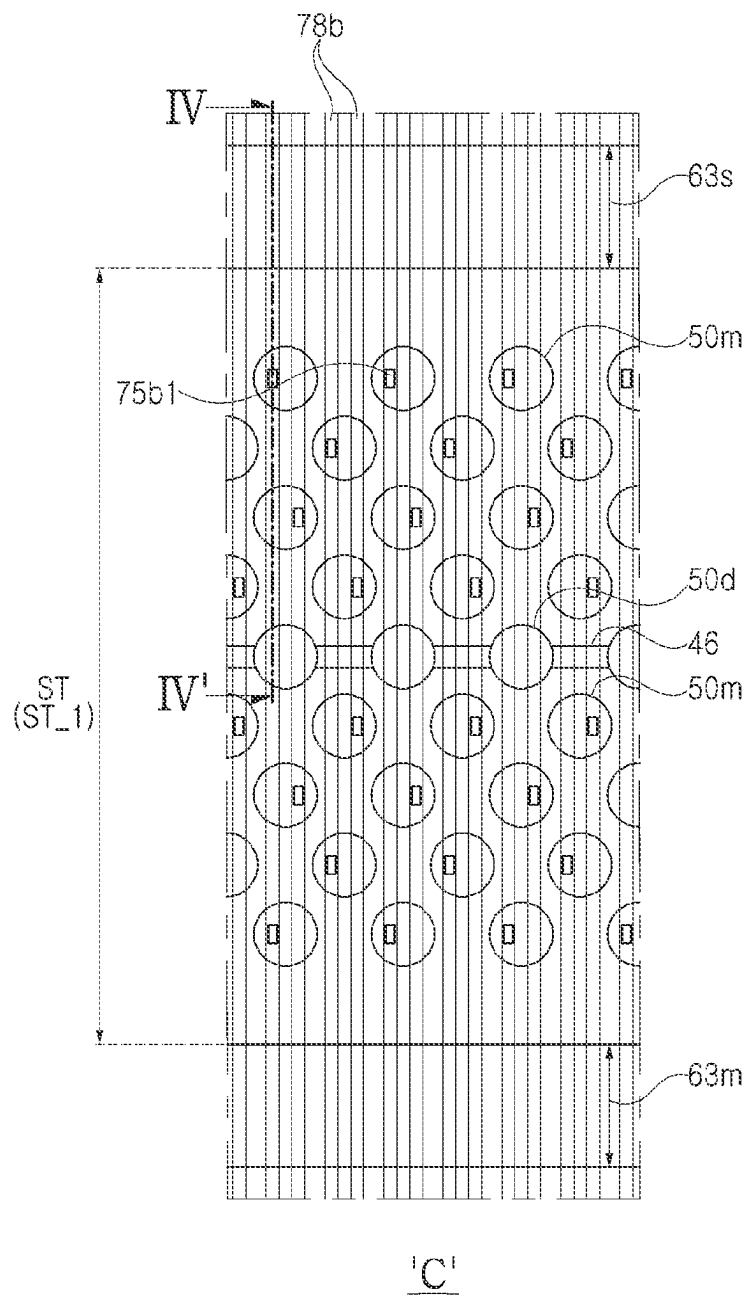
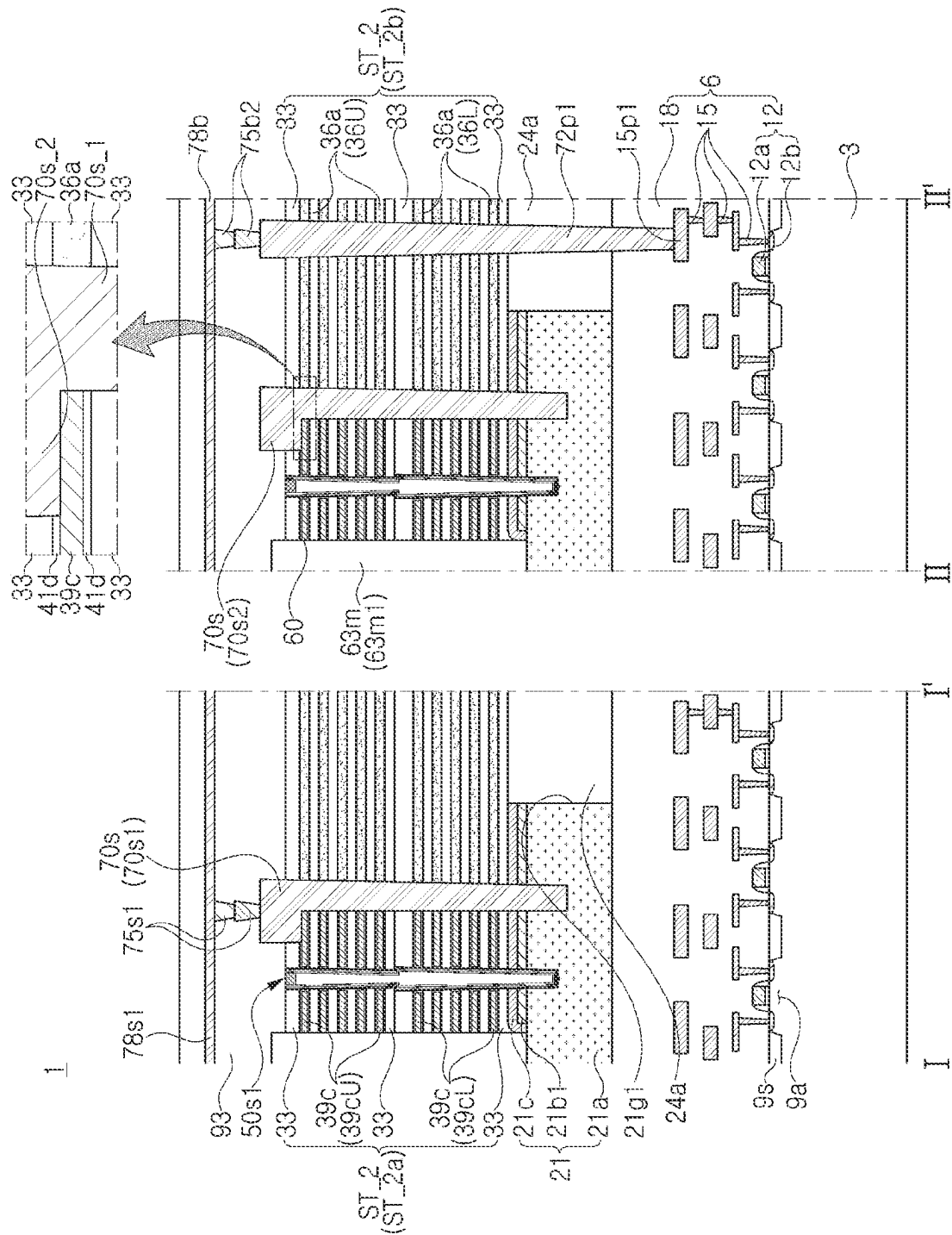


FIG. 3B



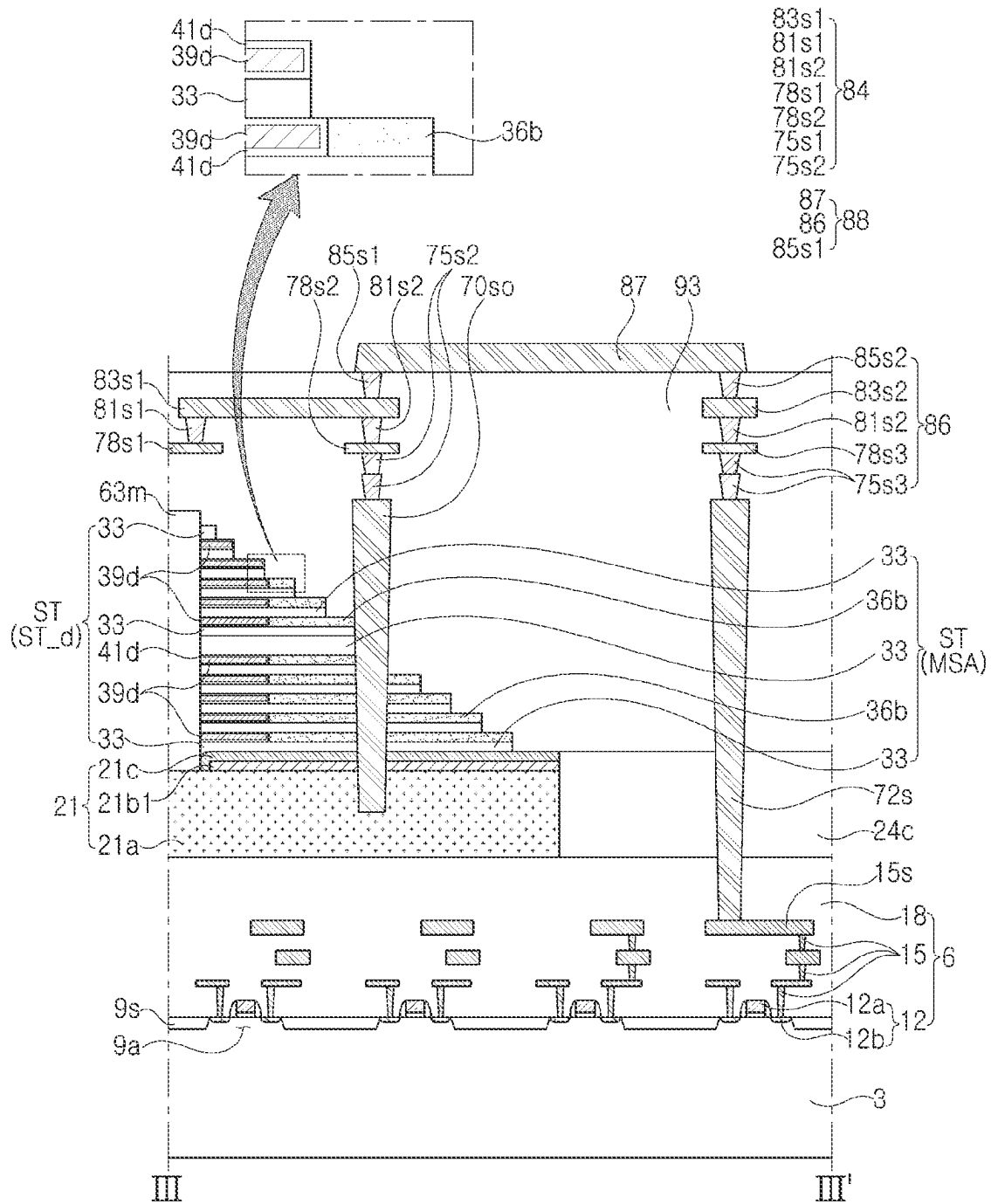
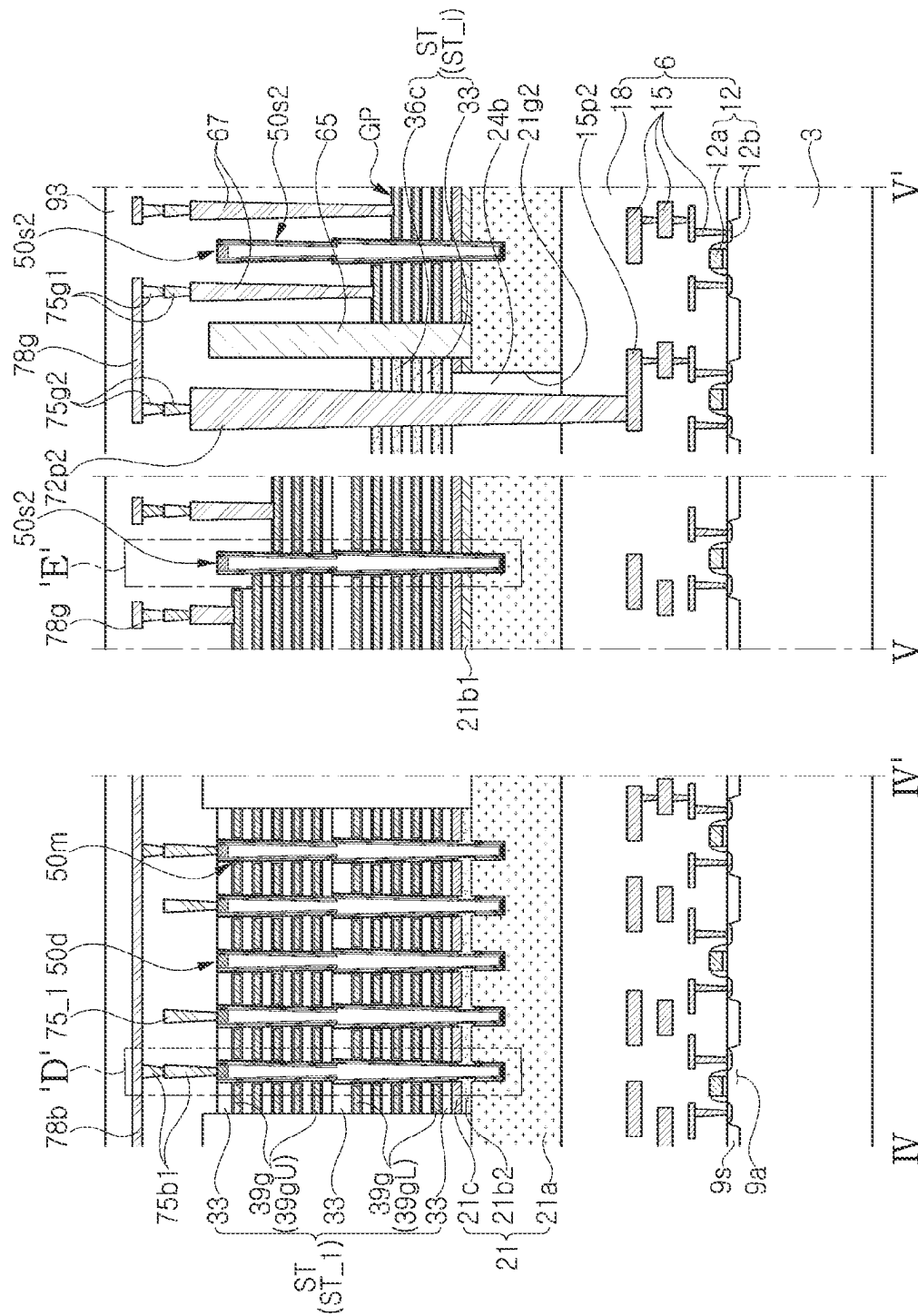


FIG. 4B

40
G²
F

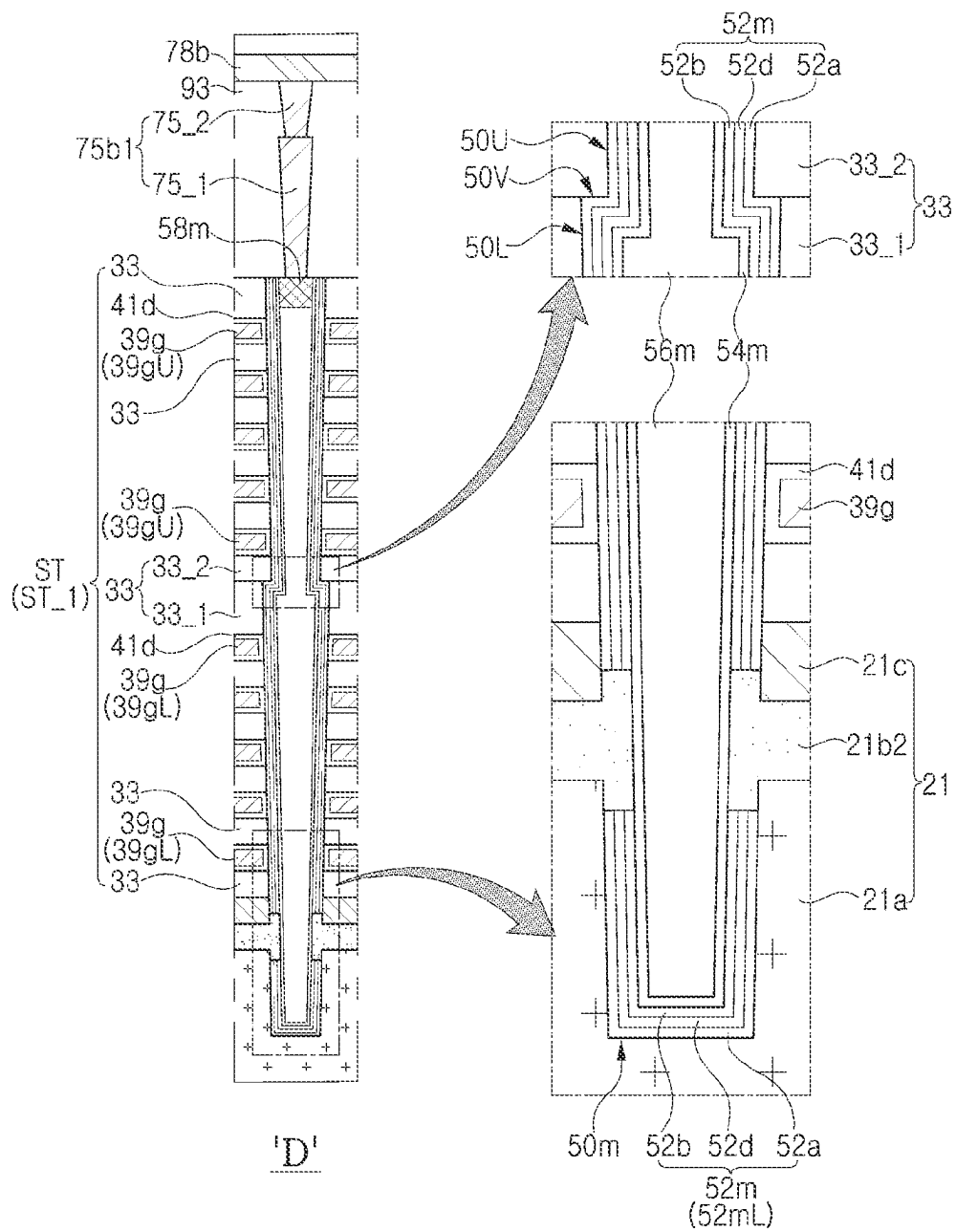


FIG. 5A

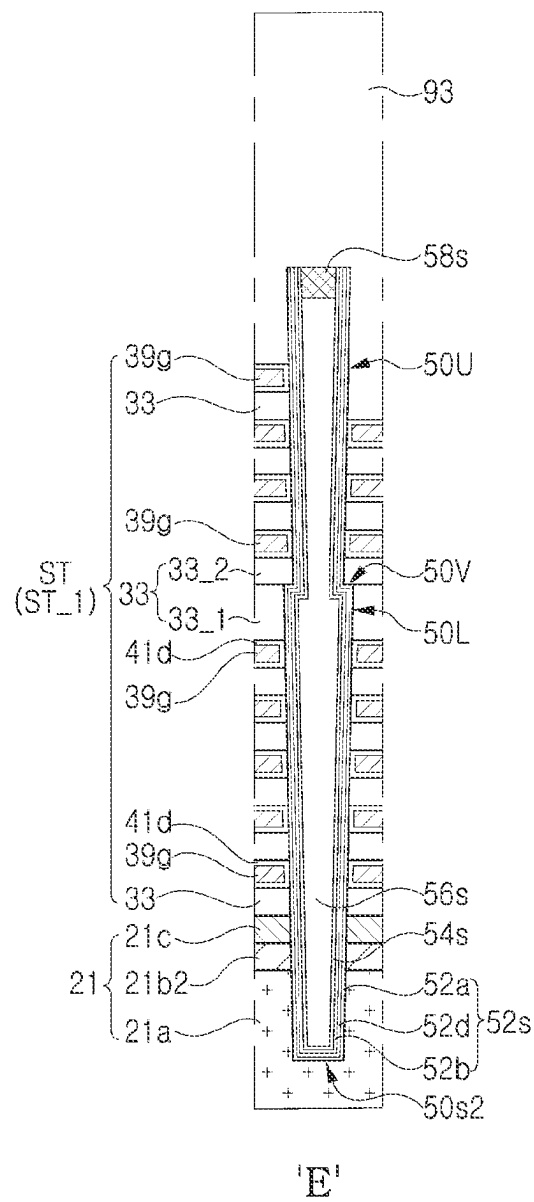
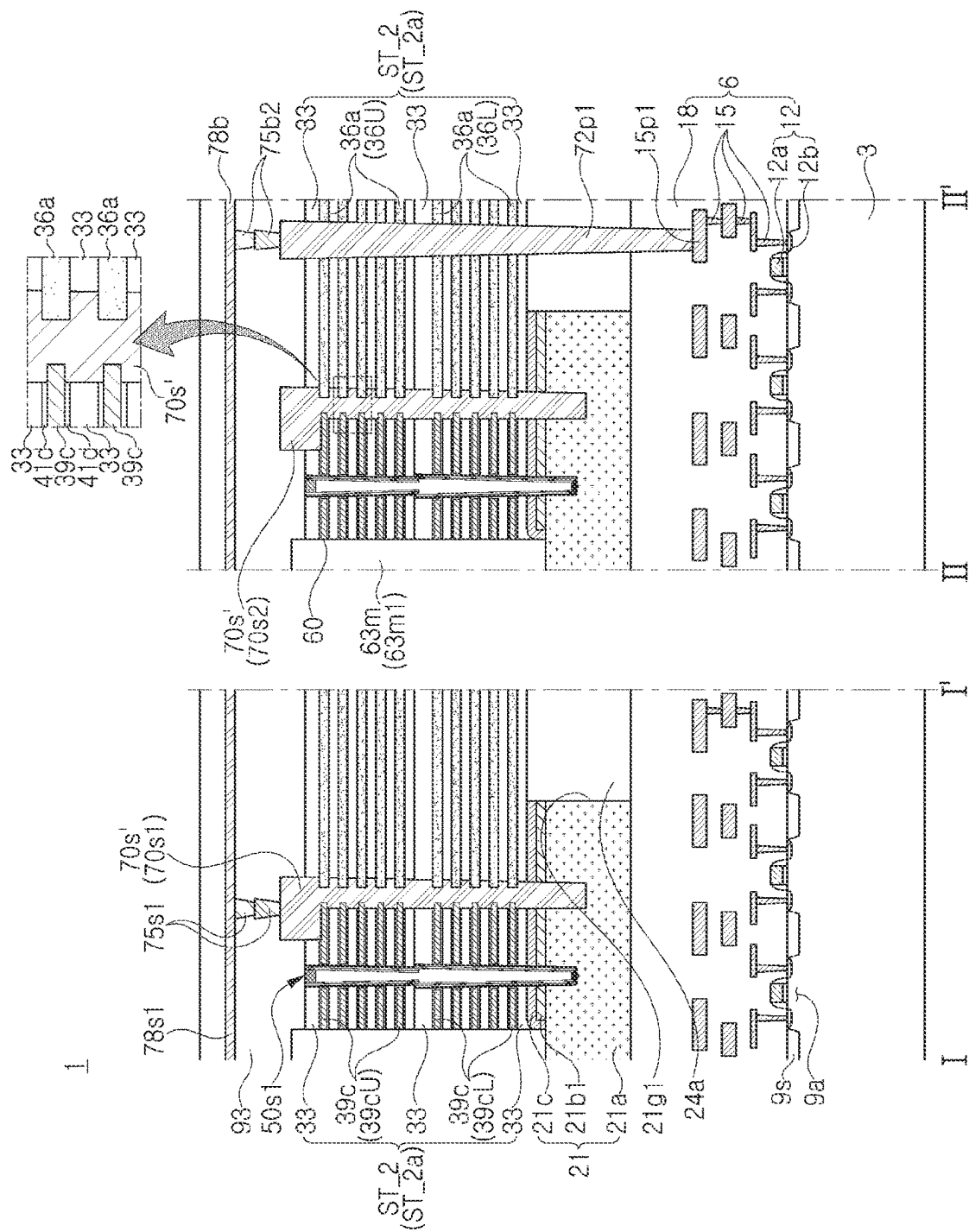


FIG. 5B



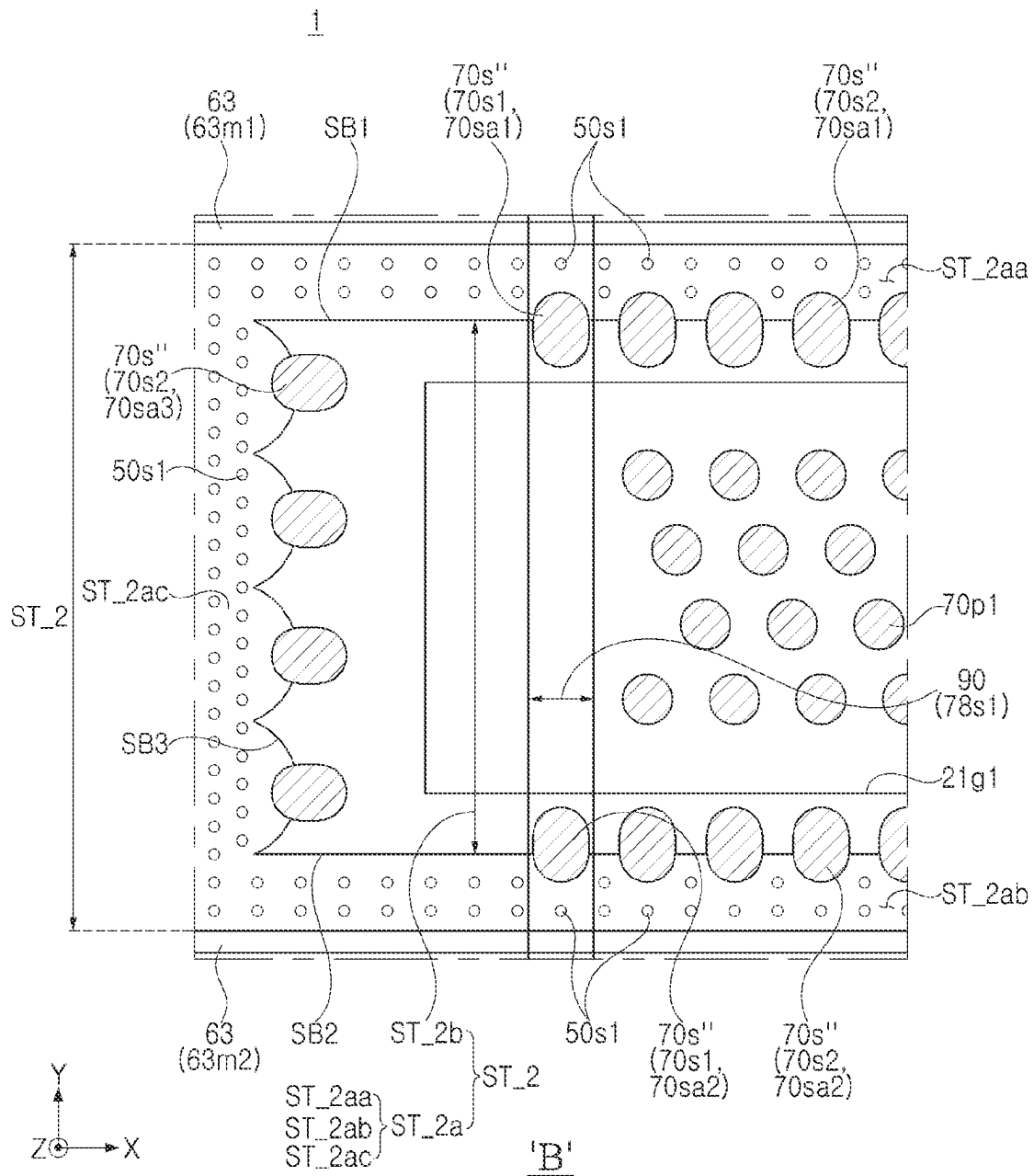
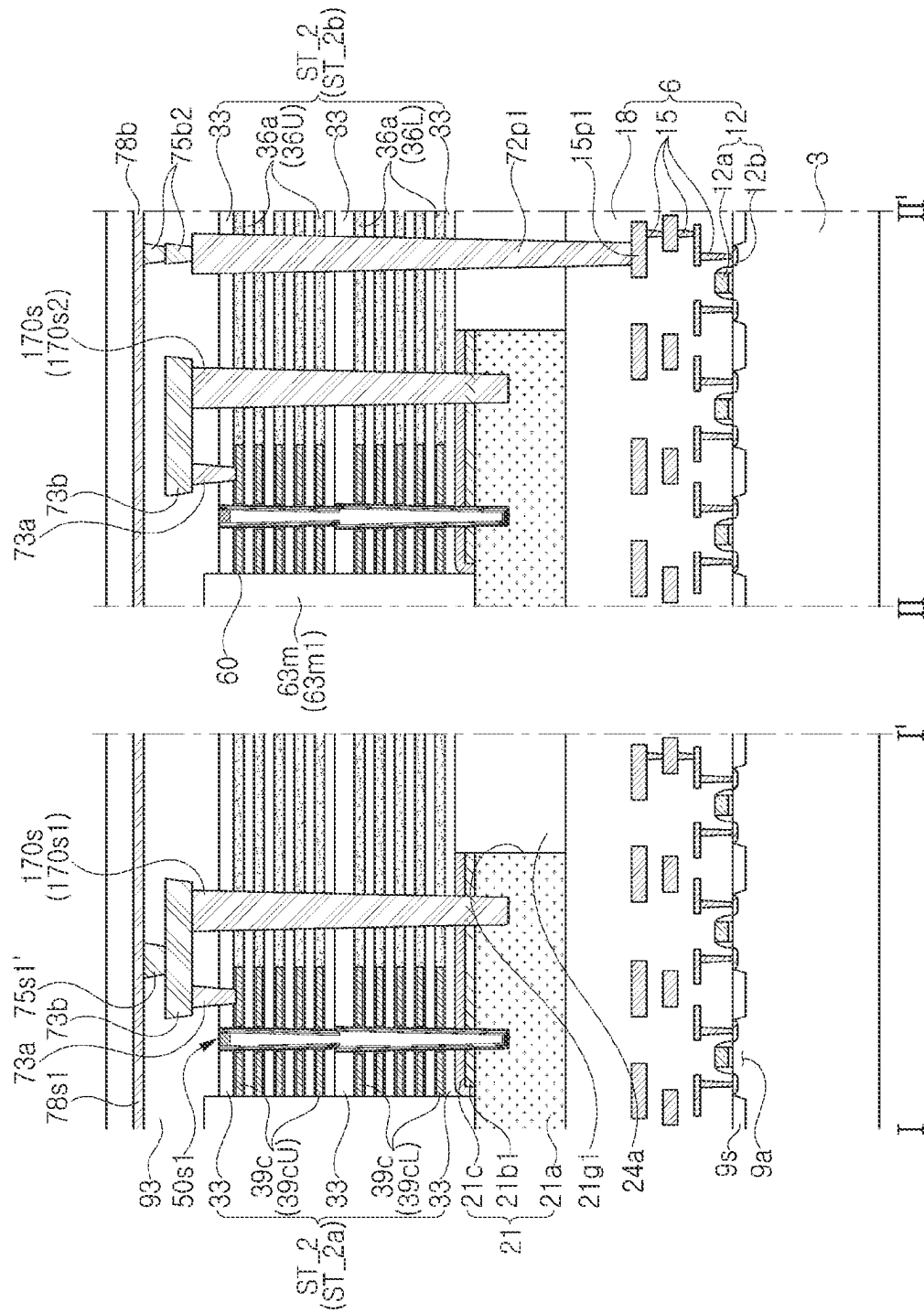


FIG. 7

[illegible]

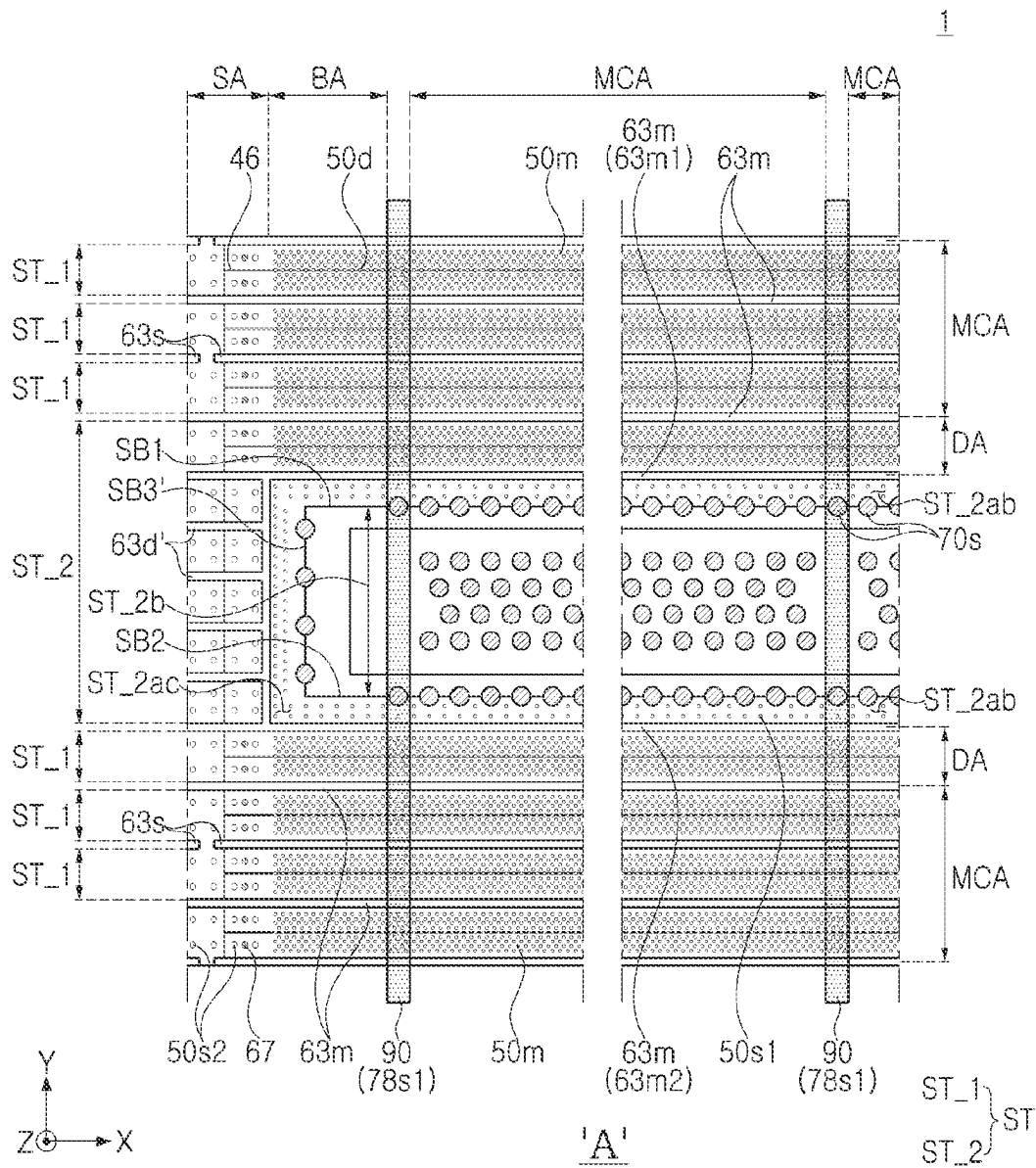


FIG. 9

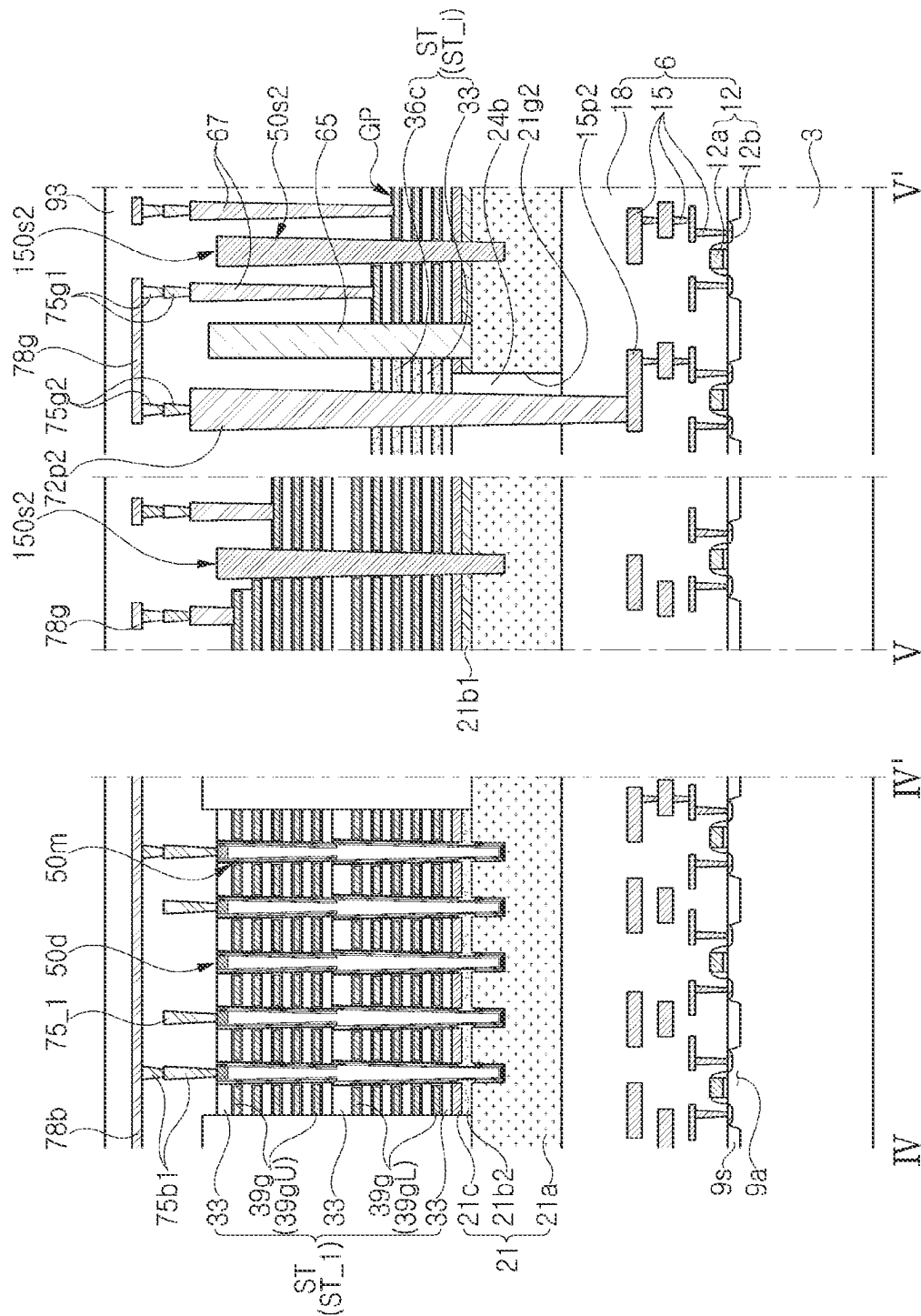


FIG. 10

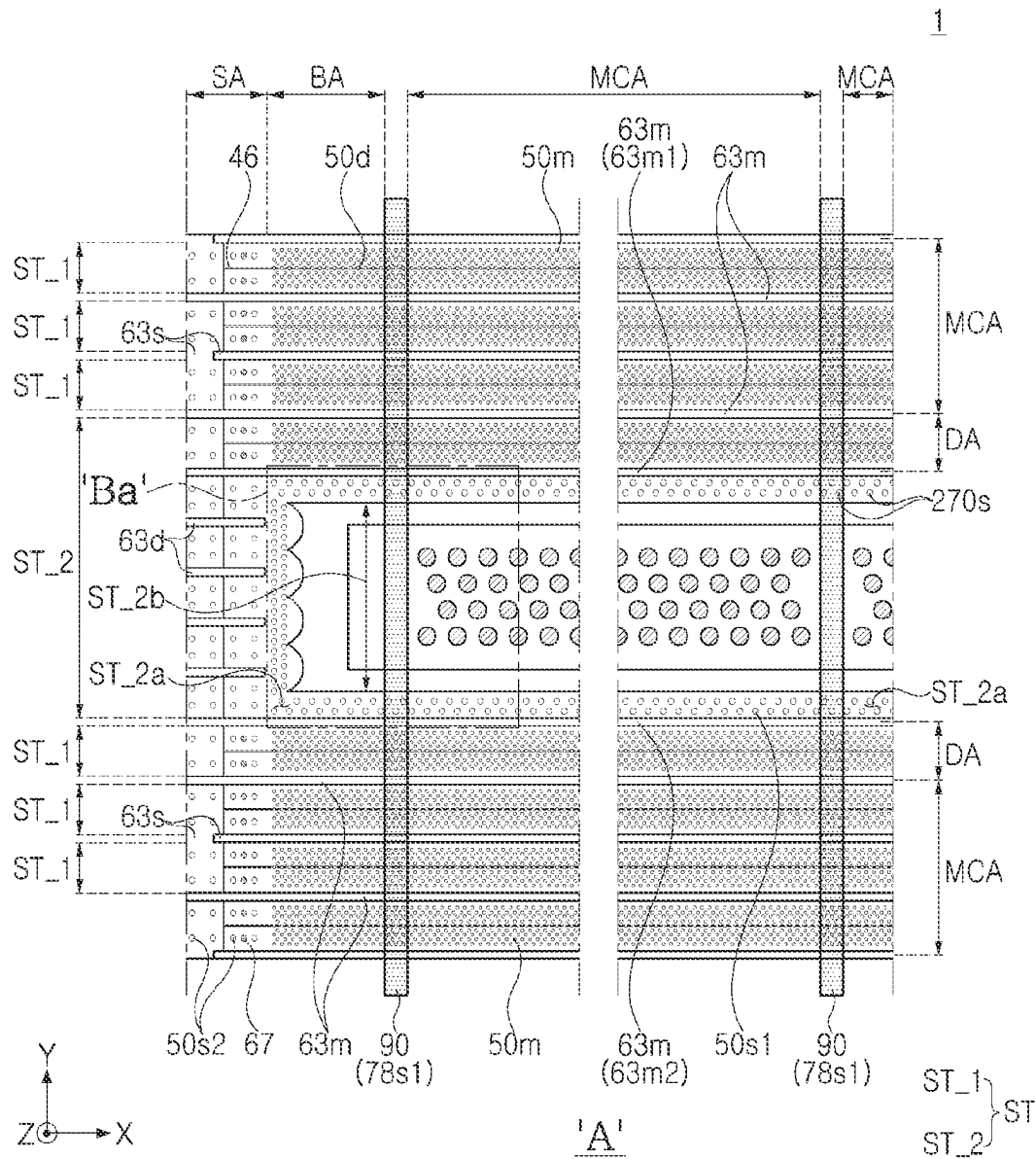


FIG. 11

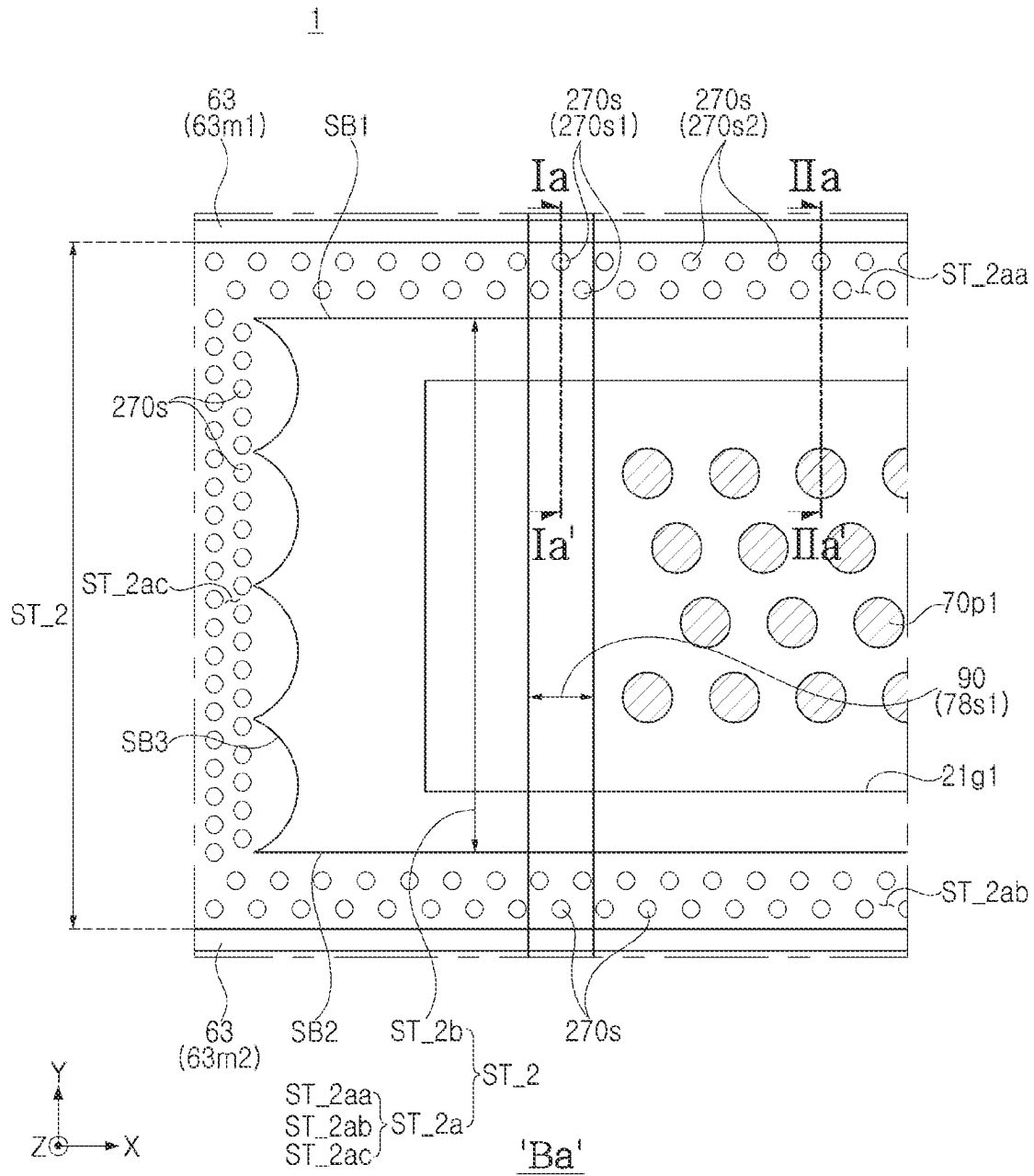


FIG. 12

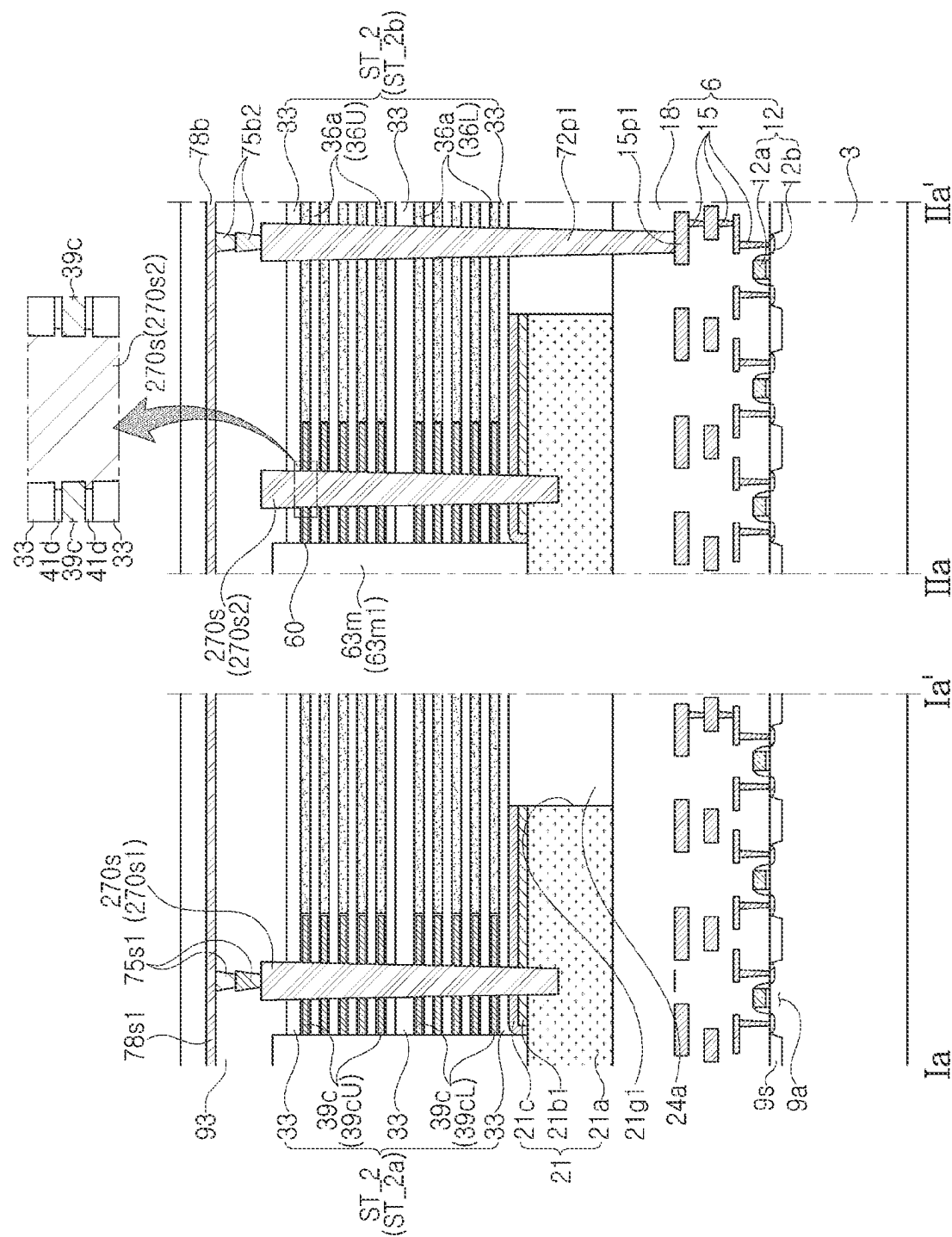


FIG. 13

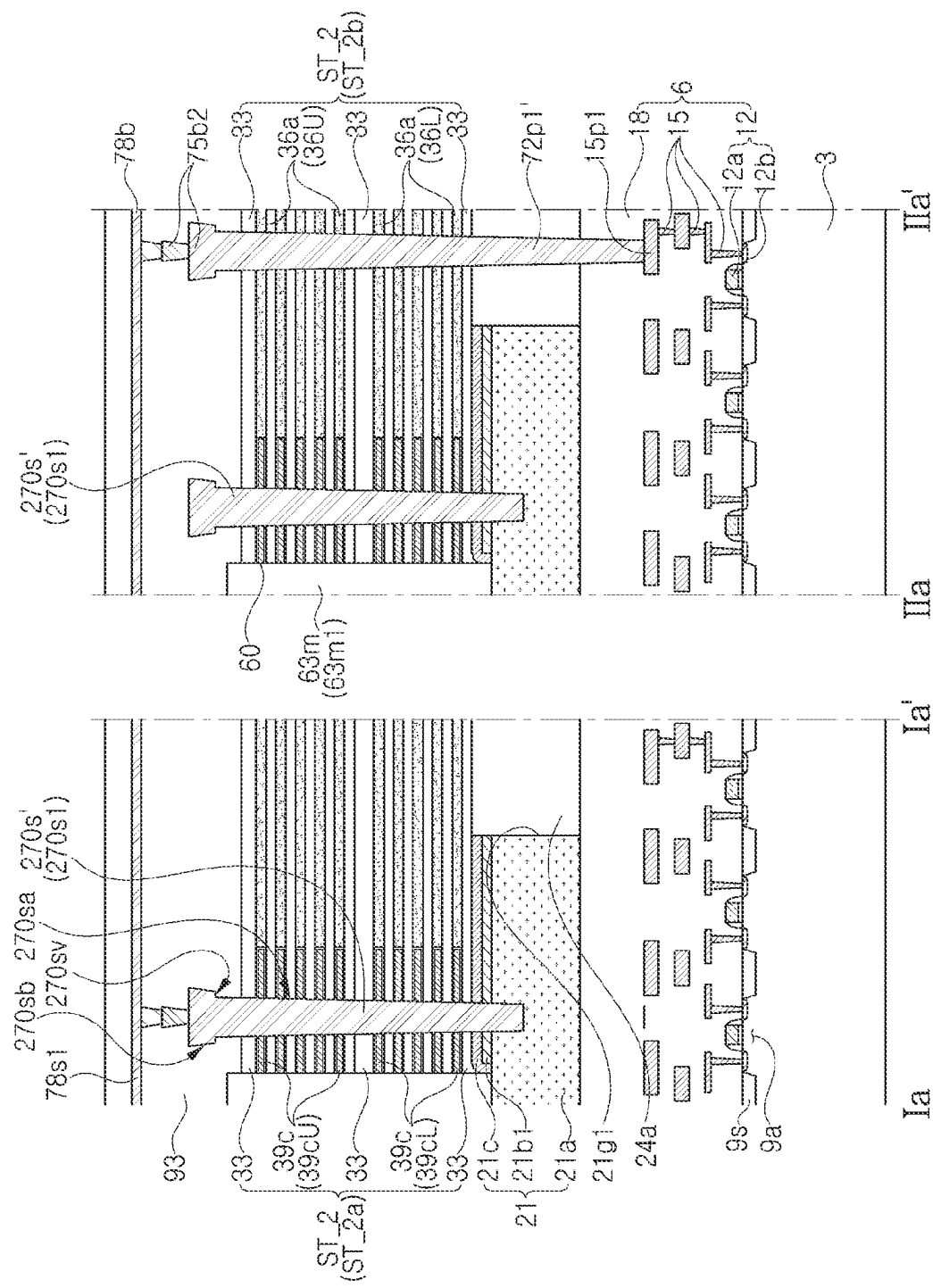


FIG. 14

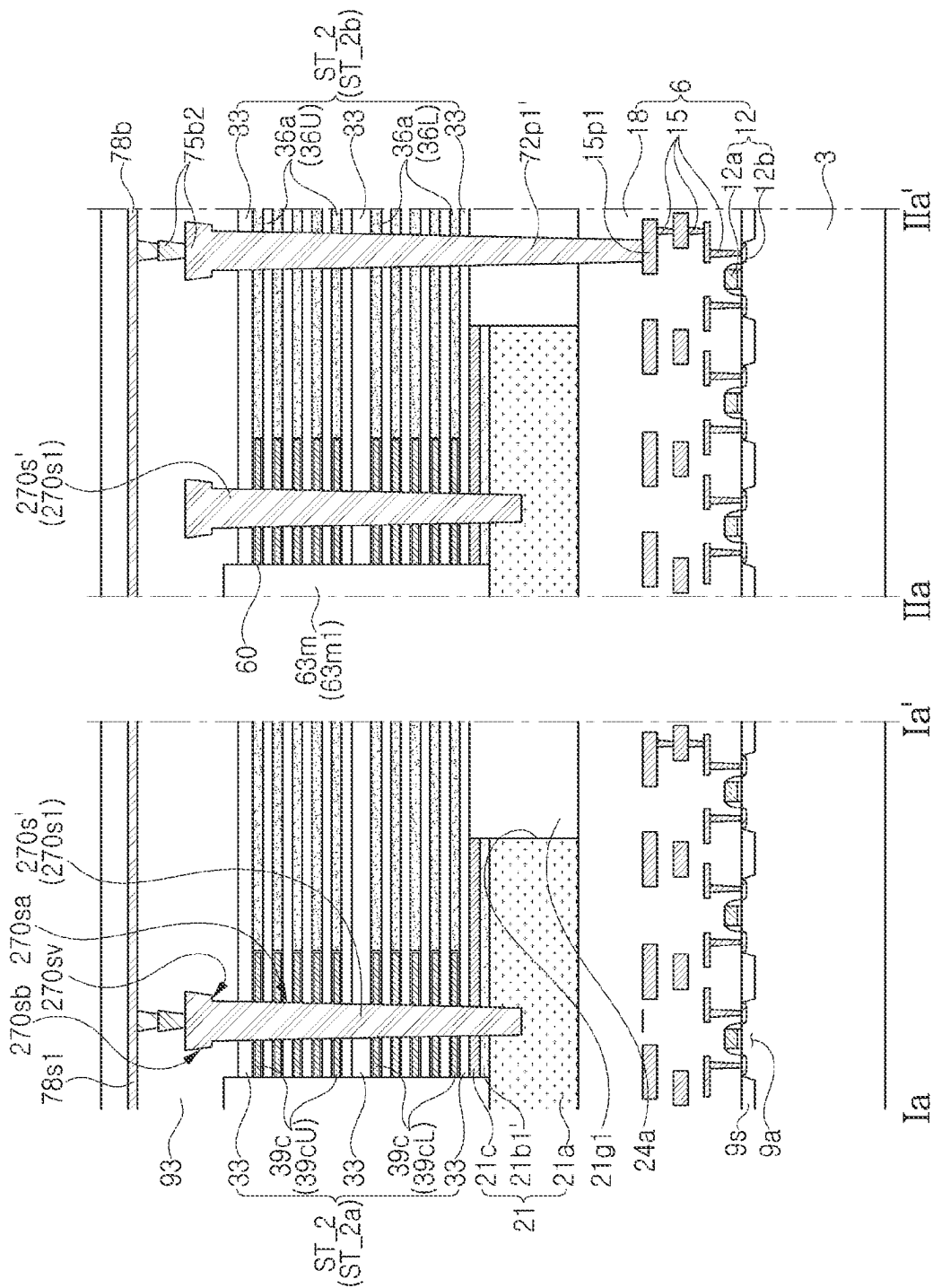


FIG. 15

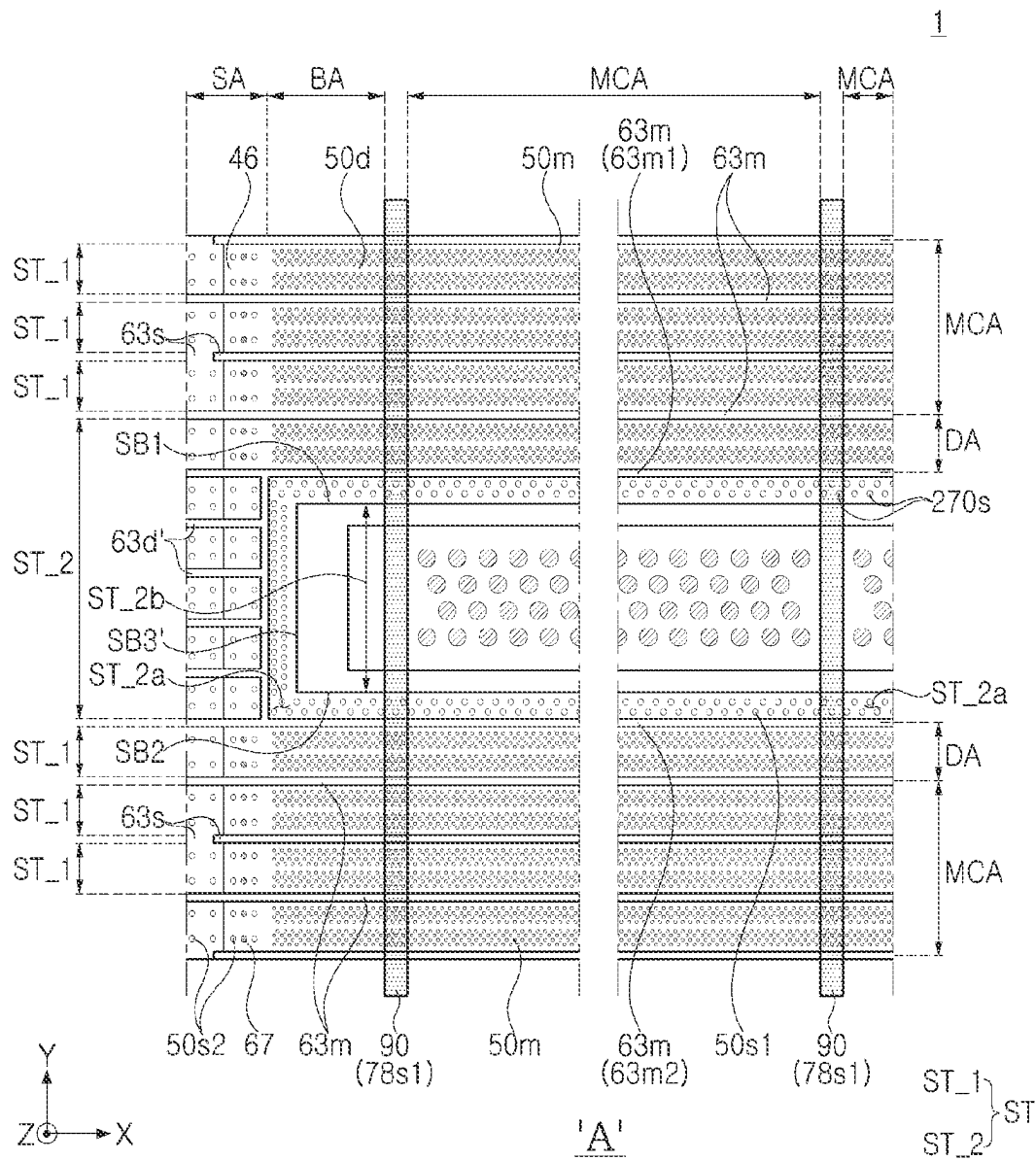


FIG. 16

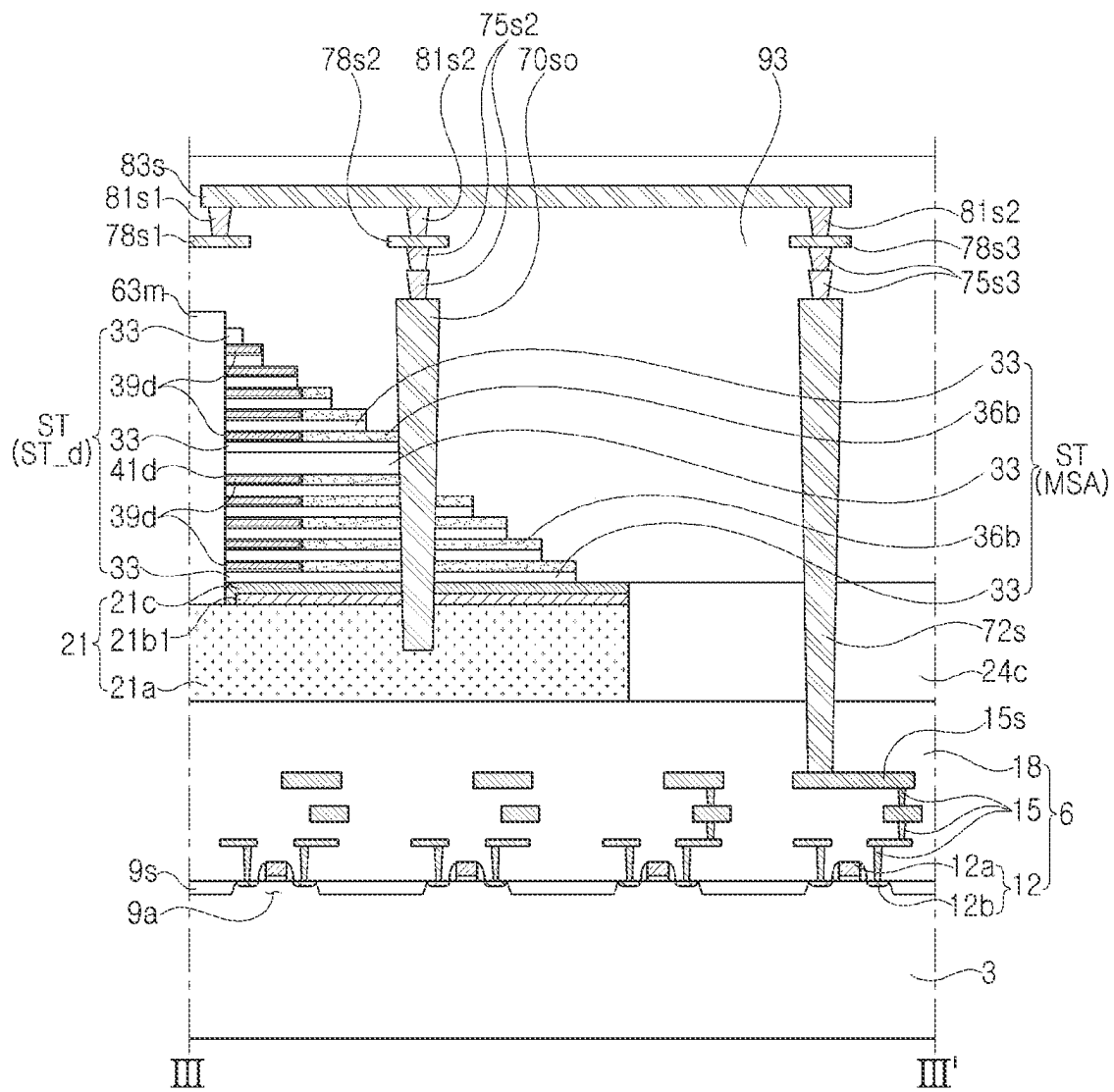


FIG. 17

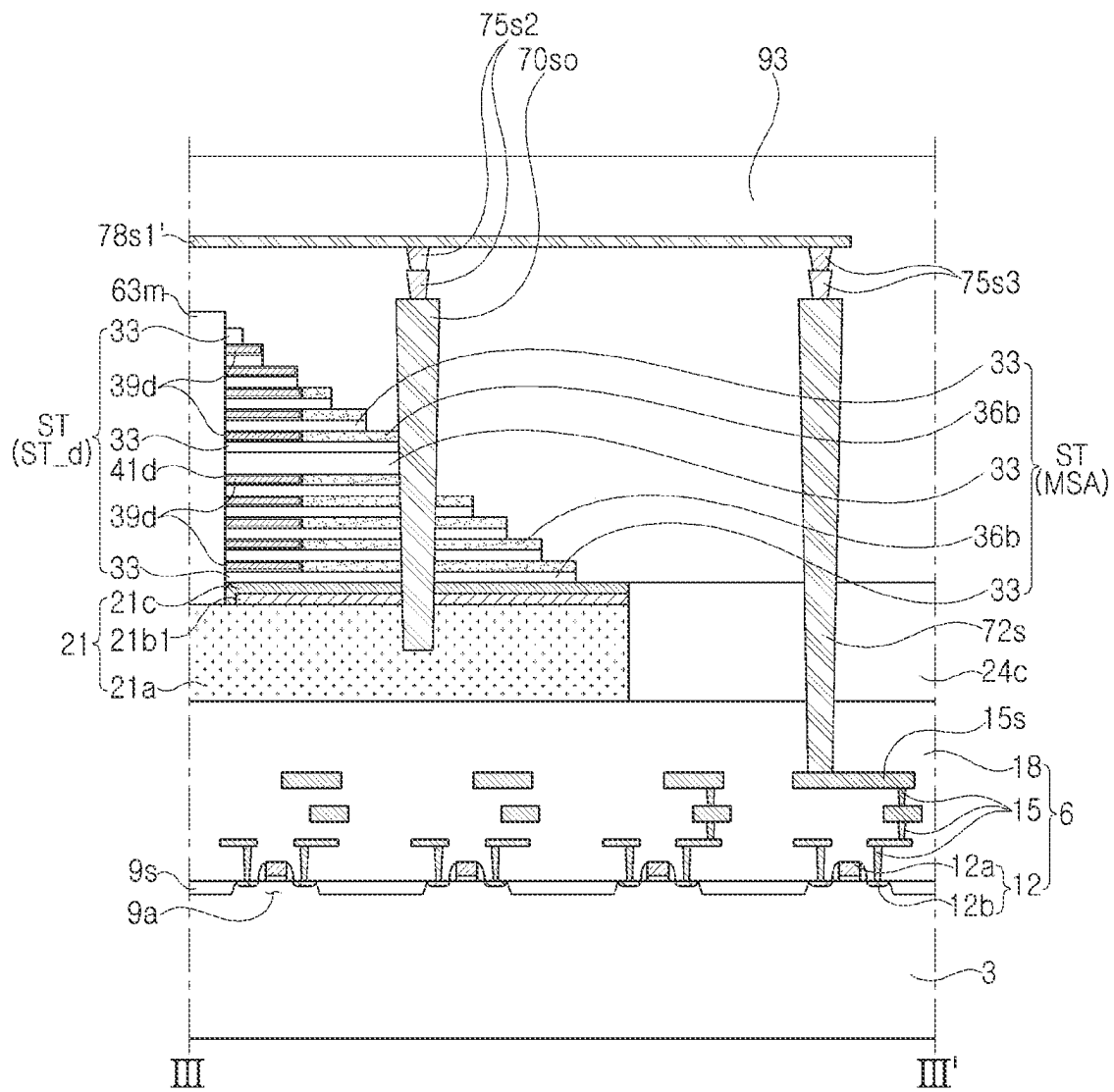


FIG. 18

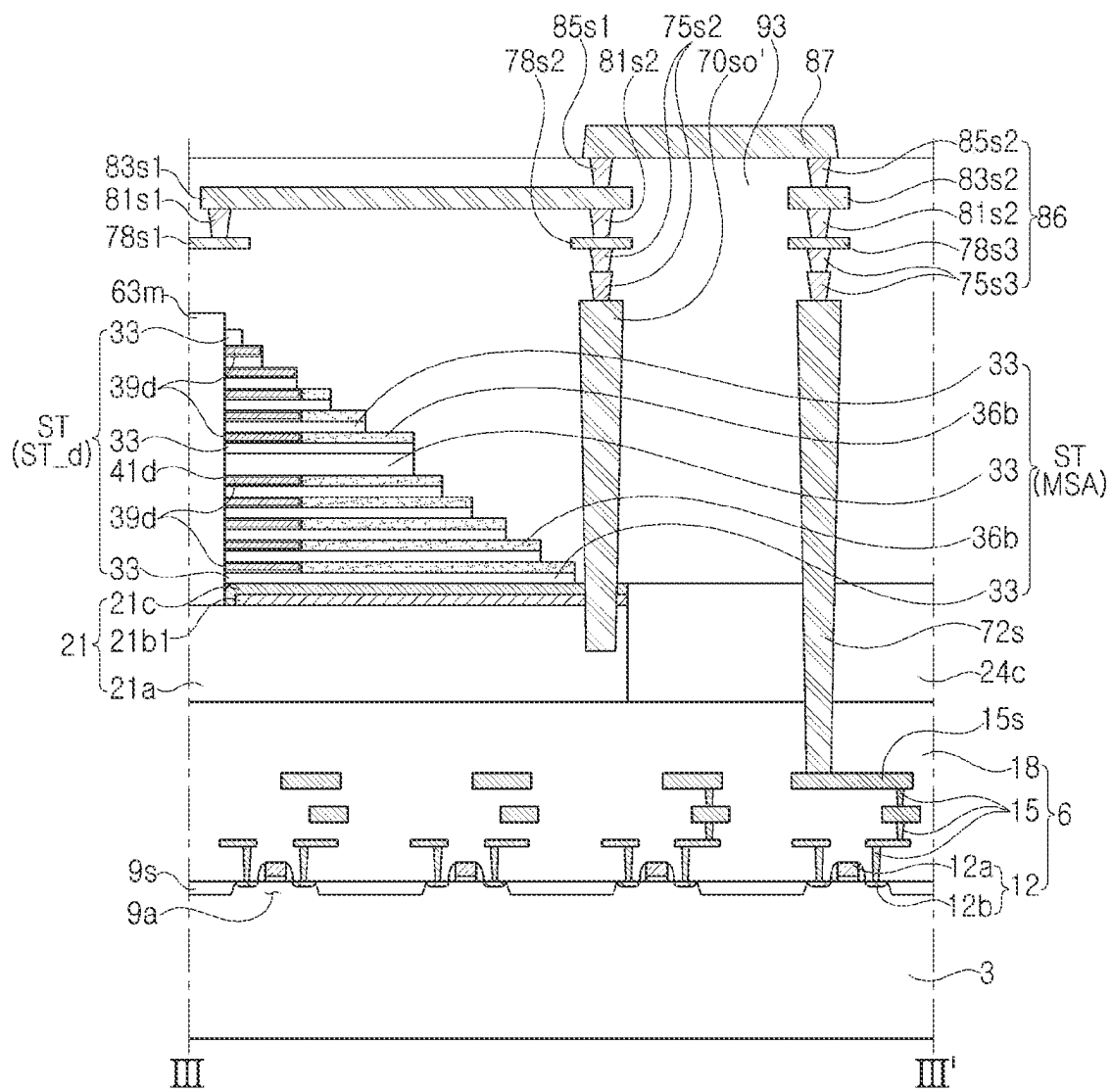


FIG. 19

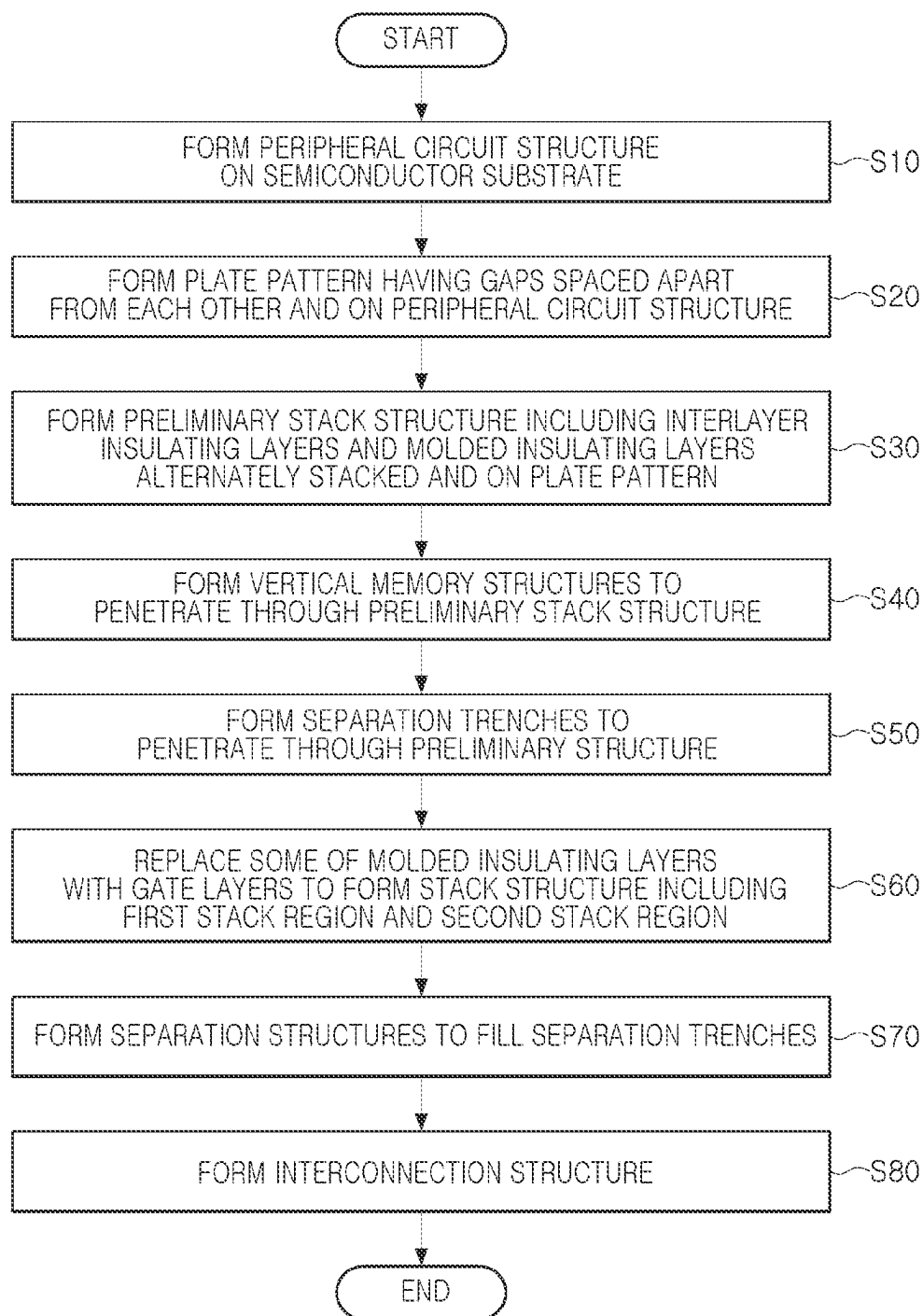


FIG. 20

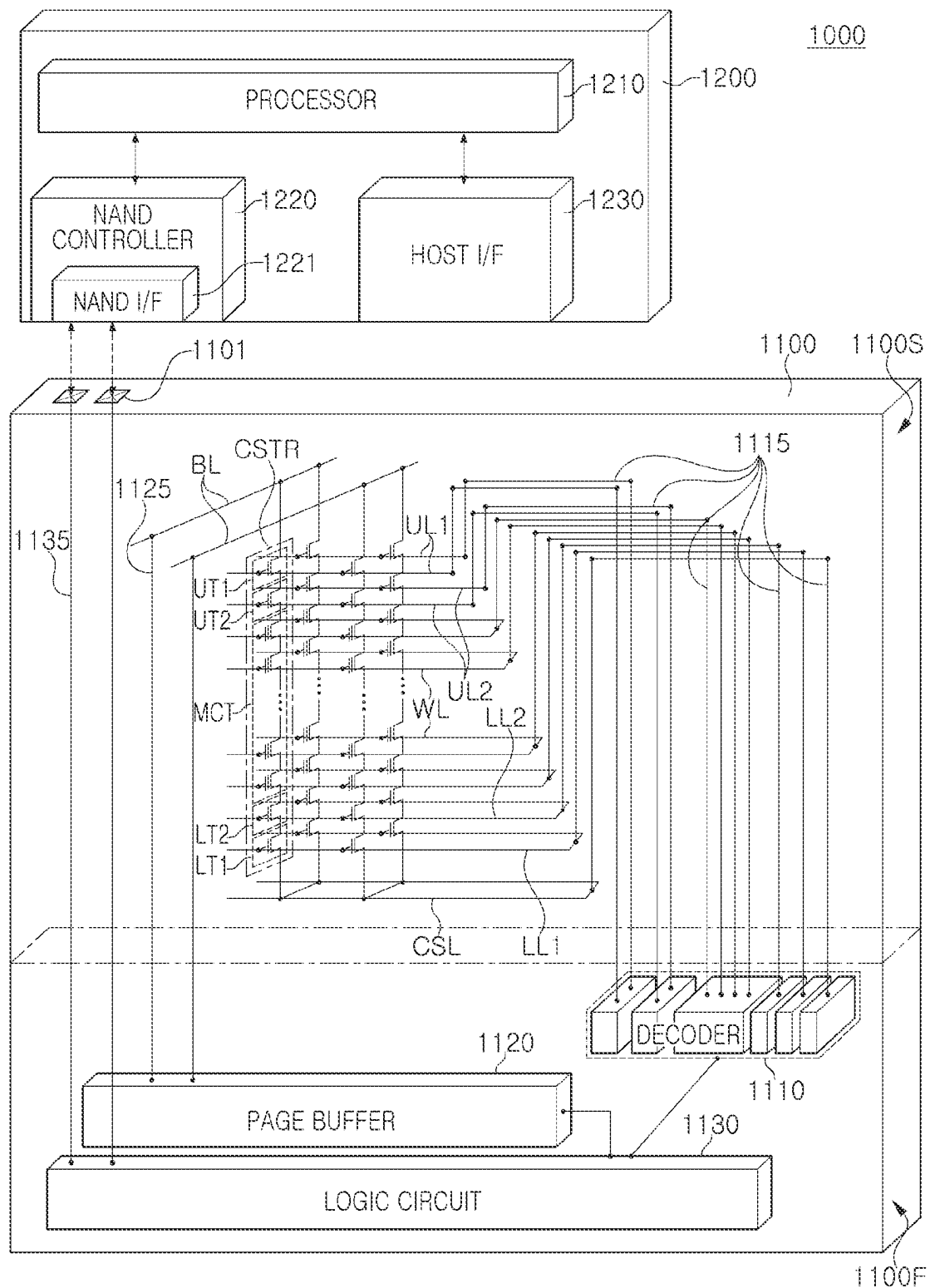


FIG. 21

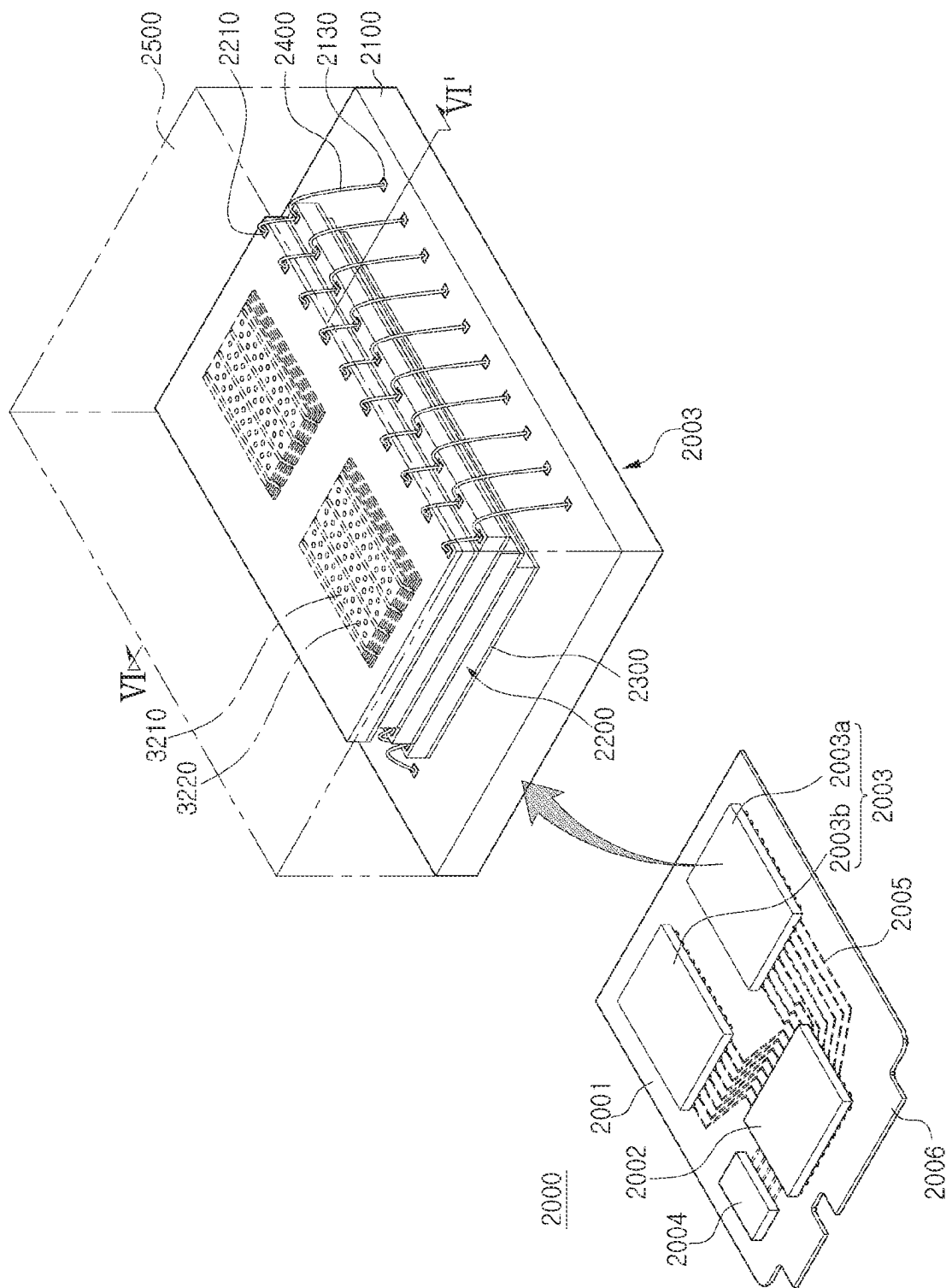


FIG. 22

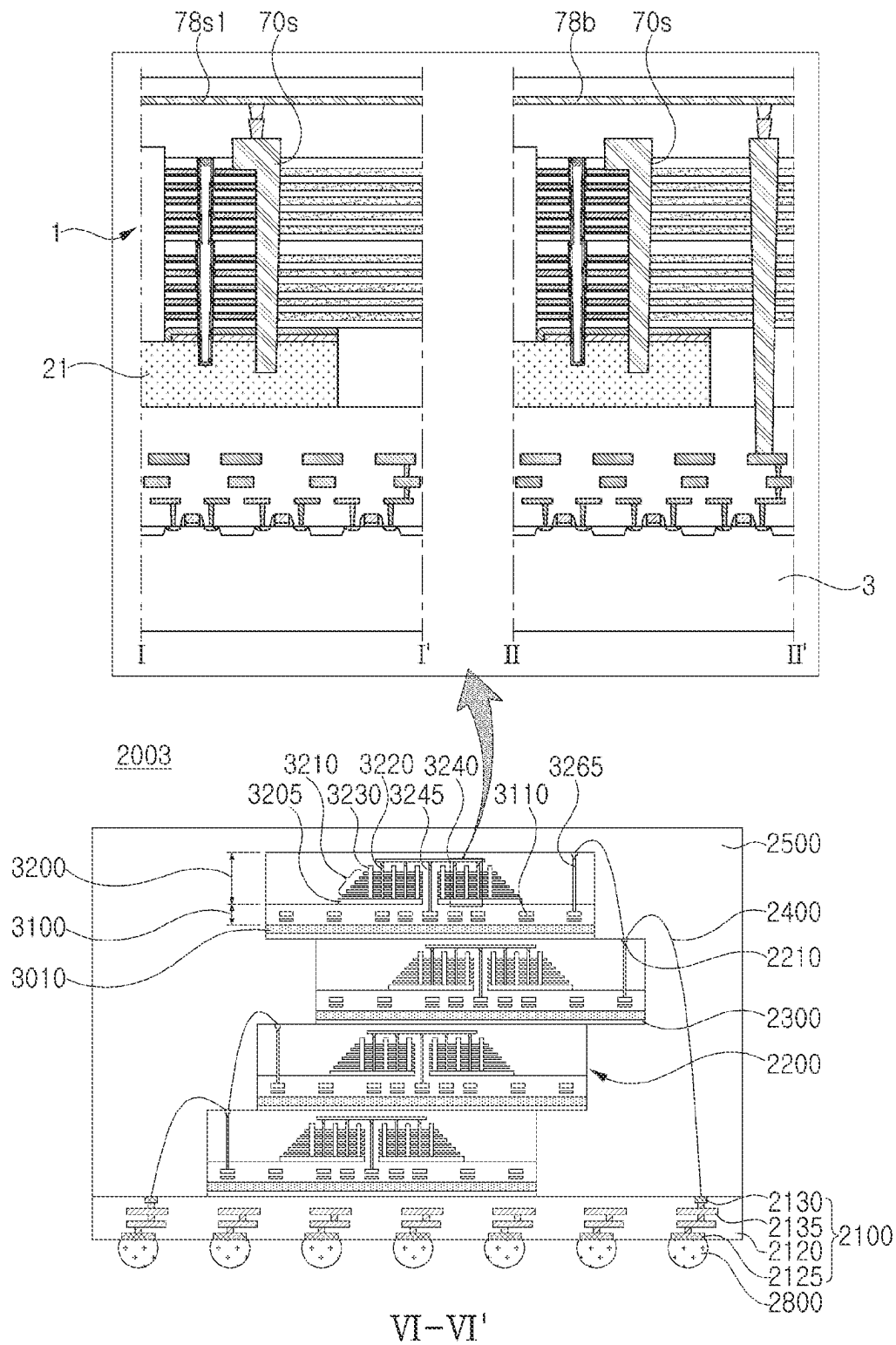


FIG. 23

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SEMICONDUCTOR DEVICES AND DATA STORAGE SYSTEMS INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims benefit of priority to Korean Patent Application No. 10-2021-0120940 filed on Sep. 10, 2021, in the Korean Intellectual Property Office, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD

The present disclosure relates to semiconductor devices and to data storage systems including the same.

BACKGROUND

There has been increasing demand for semiconductor devices that store high-capacity data for use in electronic systems that require data storage capabilities. Accordingly, there have been studies into measures for increasing the data storage capacity of semiconductor devices. For example, one proposed method for increasing data storage capacity of a semiconductor device uses a semiconductor device including memory cells arranged three-dimensionally instead of memory cells arranged two-dimensionally.

SUMMARY

Some aspects of the present disclosure provide semiconductor devices exhibiting improved electrical performance and/or having increased integration density. Some aspects of the present disclosure provide data storage systems that include semiconductor devices exhibiting improved electrical performance and/or having increased integration density.

According to some example embodiments, a semiconductor device may include: a semiconductor substrate; a peripheral circuit structure on the semiconductor substrate; a plate pattern on the peripheral circuit structure and having a gap; and a stack structure on the plate pattern and including a first stack region and a second stack region spaced apart from the first stack region. The first stack region may include gate electrodes stacked in a vertical direction that is perpendicular to an upper surface of the semiconductor substrate, and the second stack region may include both a conductor stack region including conductive layers stacked in the vertical direction and an insulator stack region including molded insulating layers at substantially the same height level as the conductive layers. The semiconductor device may also include a vertical memory structure that extends through the first stack region; and source contact plugs electrically connected to at least one of the conductive layers of the conductor stack region and contacting the plate pattern.

According to some example embodiments, a semiconductor device may include: a semiconductor substrate; a peripheral circuit structure on the semiconductor substrate; a plate pattern on the peripheral circuit structure and having gaps spaced apart from each other; and a stack structure on the plate pattern and including first stack regions spaced apart from each other and second stack regions spaced apart from each other. Each of the first stack regions may include gate electrodes stacked in a vertical direction perpendicular to an upper surface of the semiconductor substrate, and each of the second stack regions may include both a conductor stack region including conductive layers stacked in the vertical

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direction, and an insulator stack region including molded insulating layers at substantially the same height level as the conductive layers. The semiconductor device may include vertical memory structures that extend through the first stack regions; source contact plugs that extend through the second stack regions; separation structures that extend through the stack structure; source connection lines having a higher level than the stack structure; and bitlines having a higher level than the stack structure. When viewed in a plan view, each of the gaps may have a line shape or a bar shape that extends in a first direction that is parallel to the upper surface of the semiconductor substrate. When viewed in the plan view, the first and second stack regions are spaced apart from each other in a second direction that is parallel to the upper surface of the semiconductor substrate and perpendicular to the first direction. The insulator stack regions of the second stack regions may overlap the gaps. In each of the second stack regions, at least one of the conductive layers of the second stack region may be electrically connected to the source contact plugs that extend through the second stack region. Each of the source connection lines may have a line shape that extends in the second direction. Each of the bitlines may have a line shape that extends in the second direction. The source contact plugs may include first source contact plugs that overlap the source connection lines, and second source contact plugs that overlap the bitlines.

According to some example embodiments, a data storage system may include: a semiconductor device including an input/output pattern; and a controller electrically connected to the semiconductor device through the input/output pattern and configured to control the semiconductor device. The semiconductor device may include: a semiconductor substrate; a peripheral circuit structure on the semiconductor substrate; a plate pattern on the peripheral circuit structure and having gaps spaced apart from each other; and a stack structure on the plate pattern and including first stack regions spaced apart from each other and second stack regions spaced apart from each other. Each of the first stack regions may include gate electrodes stacked in a vertical direction that is perpendicular to an upper surface of the semiconductor substrate, and each of the second stack regions may include both a conductor stack region including conductive layers stacked in the vertical direction, and an insulator stack region including molded insulating layers at substantially the same height level as the conductive layers. The semiconductor device may also include vertical memory structures that extend through the first stack regions; source contact plugs that extend through the second stack regions; separation structures that extend through the stack structure; source connection lines having a higher level than the stack structure; and bitlines having a higher level than the stack structure. When viewed in a plan view, each of the gaps may have a line shape or a bar shape that extends in a first direction that is parallel to the upper surface of the semiconductor substrate. When viewed in the plan view, the first and second stack regions may be spaced apart from each other in a second direction that is parallel to the upper surface of the semiconductor substrate and perpendicular to the first direction. The insulator stack regions of the second stack regions may overlap the gaps. In each of the second stack regions, at least one of the conductive layers of the second stack region may be electrically connected to the source contact plugs penetrating through the second stack region. Each of the source connection lines may have a line shape that extends in the second direction. Each of the bitlines may have a line shape that extends in the second direction. The source contact plugs may include first source

contact plugs that overlap the source connection lines, and second source contact plugs that overlap the bitlines.

BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings.

FIGS. 1 to 5B are diagrams illustrating an example of a semiconductor device according to some example embodiments.

FIG. 6 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 7 is plan view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 8 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 9 is plan view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 10 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIGS. 11 to 13 are diagrams illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 14 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 15 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 16 is plan view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 17 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 18 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 19 is a cross-sectional view illustrating a modified example of a semiconductor device according to some example embodiments.

FIG. 20 is a process flowchart illustrating an example of a method of forming a semiconductor device according to some example embodiments.

FIG. 21 is a schematic diagram illustrating a data storage system including a semiconductor device according to some example embodiments.

FIG. 22 is a schematic perspective view illustrating a data storage system including a semiconductor device according to some example embodiments.

FIG. 23 is a schematic cross-sectional view illustrating a data storage system including a semiconductor device according to some example embodiments.

DETAILED DESCRIPTION

Hereinafter, some example embodiments will be described with reference to the accompanying drawings. Hereinafter, terms such as “up,” “upper portion,” “upper surface,” “down,” “lower portion,” “lower surface,” “side

surface,” etc., may be understood with reference to the accompanying drawings. Terms such as “upper,” “middle,” and “lower” may be replaced with other terms, e.g., “first,” “second,” and “third,” etc., to be used to describe elements of the specification. Terms such as “first” and “second” may be used to describe various elements, but the elements are not limited by the terms, and a “first element” may be referred to herein interchangeably as a “second element,” as appropriate.

An example of a semiconductor device according to some example embodiments will be described with reference to FIGS. 1, 2, 3A, 3B, 4A, 4B, and 4C. FIG. 1 is plan view illustrating an example of a semiconductor device according to some example embodiments, and FIG. 2 is an enlarged plan view of region “A” of FIG. 1, FIG. 3A is an enlarged plan view of region “B” of FIG. 2, FIG. 3B is an enlarged plan view of region “C” of FIG. 2, FIG. 4A is a cross-sectional view illustrating regions taken along lines I-I' and II-II' of FIG. 3A, FIG. 4B is a cross-sectional view illustrating regions taken along line of FIG. 1, and FIG. 4C is a cross-sectional view illustrating regions taken along line IV-IV' of FIG. 3B and line V-V' of FIG. 1.

Referring to FIGS. 1 to 4C, a semiconductor device 1 according to some example embodiments may include a semiconductor substrate 3, a peripheral circuit structure 6 on the semiconductor substrate 3, a stack structure ST including a plate pattern 21 having gaps 21g1 and 21g2, and a first stack region ST_1 and a second stack region ST_2, on the plate pattern 21 and spaced apart from each other, on the peripheral circuit structure 6, Vertical memory structures 50m may penetrate or extend through at least the stack structure ST and may contact the plate pattern 21, and source contact plugs 70s may penetrate or extend through the stack structure ST and may contact the plate pattern 21.

The semiconductor device 1 according to some example embodiments may further include first and second vertical support structures 50s1 and 50s2 and vertical dummy structures 50d that may penetrate or extend through at least the stack structure ST and may be in contact with the plate pattern 21. The first and second vertical support structures 50s1 and 50s2 and the vertical dummy structures 50d may be dummy structures electrically insulated from a bitline 78b. The first and second vertical support structures 50s1 and 50s2 and the vertical dummy structures 50d may serve as supports to prevent collapse or deformation of the stacked structure ST. The peripheral circuit structure 6 may include a peripheral circuit 12 on the semiconductor substrate 3, a peripheral interconnection structure 15 electrically connected to the peripheral circuit 12 on the semiconductor substrate 3, and a lower insulating structure 18 covering the peripheral circuit 12 and the peripheral interconnection structure 15 on the semiconductor substrate 3. The peripheral circuit 12 may include a peripheral transistor including source/drain regions 12b and a peripheral gate 12a. The source/drain regions 12b may be in an active region 9a defined by an isolation region 9s on the semiconductor substrate 3. The peripheral gate 12a may be on the active region 9a between the source/drain regions 12b. The peripheral interconnection structure 15 may include a first peripheral pad 15p1, a second peripheral pad 15p2, and a source pad 15s.

The plate pattern 21 may include a silicon layer, for example, a polysilicon layer. At least a portion of the plate pattern 21 may be a common source in a flash memory device.

The plate pattern 21 may include a lower layer 21a, a first intermediate layer 21b1 and a second intermediate layer

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21b2 at the same height level on the lower layer **21a**, and an upper layer **21c** that covers the first and second intermediate layers **21b1** and **21b2** on the lower layer **21a**.

The lower layer **21a** may include a doped polysilicon layer, for example, a polysilicon layer having a first conductivity type (e.g., N-type conductivity). The first intermediate layer **21b1** may include a single layer or a plurality of layers. For example, the first intermediate layer **21b1** may include a silicon oxide layer, a silicon nitride layer, and a silicon oxide layer stacked in order. The second intermediate layer **21b2** may include a doped polysilicon layer, for example, a polysilicon layer having the first conductivity type (e.g., N-type conductivity). The upper layer **21c** may include a doped polysilicon layer, for example, a polysilicon layer having the first conductivity type (e.g., N-type conductivity). The first intermediate layer **21b1** may be spaced apart from separation structures **63m**, **63s**, and **63d**, and the second intermediate layer **21b2** may be in contact with the separation structures **63m** and **63s**. A region, for example formed of polysilicon having the first conductivity type (e.g., N-type conductivity), in the plate pattern **21** may be a common source of a flash memory device.

The stacked structure **ST** may include a first stack region **ST_1** and a second stack region **ST_2**.

In the stack structure **ST**, the first stack region **ST_1** may include gate electrodes **39g** stacked in a vertical direction **Z** that is perpendicular to an upper surface of the semiconductor substrate **3**. The first stack region **ST_1** may further include interlayer insulating layers **33** repeatedly and alternately stacked with the gate electrodes **39g**. The first stack region **ST_1** may further include a dielectric layer (**41d** of FIGS. **5A** and **5B**) covering upper and lower surfaces of each of the gate electrodes **39g** and covering a portion of a side surface of each of the gate electrodes **39g**.

In the stack structure **ST**, the second stack region **ST_2** may include a conductor stack region **ST_2a**, including conductive layers **39c** stacked in the vertical direction **Z**, and an insulator stack region **ST_2b** including molded-insulating layers **36a** at substantially the same height level as the conductive layers **39c**.

The conductor stack region **ST_2a** may further include interlayer insulating layers **33** repeatedly and alternately stacked with the conductive layers **39c**. The conductor stack region **ST_2a** may further include a dielectric layer (**41d** of FIG. **4A**) covering an upper surface and a lower surface of each of the conductive layers **39c** and covering a portion of a side surface of each of the gate electrodes **39g**. The insulator stack region **ST_2b** may further include interlayer insulating layers **33** repeatedly and alternately stacked with the molded insulating layers **36a**.

The interlayer insulating layers **33** may be formed of silicon oxide. The gate electrodes **39g** and the conductive layers **39c** may be formed of the same conductive material. The gate electrodes **39g** and the conductive layers **39c** may include at least one of doped polysilicon, metal nitride (for example, titanium nitride (TiN), or the like), a metal-semiconductor compound (for example, titanium silicon (TiSi), nickel silicon (NiSi), or the like), and/or a metal (for example, tungsten (W), or the like). The dielectric layer **41d** may be formed of silicon oxide or a high-k dielectric (for example, aluminum oxide (AlO), or the like). The molded insulating layers **36a** may be formed of silicon nitride.

The stack structure **ST** may include a mold staircase region **MSA** and a dummy staircase region **ST_d** between the mold staircase region **MSA** and the first stack region **ST_1**. The dummy staircase region **ST_d** may further include interlayer insulating layers **33** and the dummy gate

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electrodes **39d**, repeatedly and alternately stacked, and a dielectric layer (**41d** of FIG. **4B**) covering an upper surface and a lower surface of each of the dummy gate electrodes **39d** and covering a portion of a side surface of each of the dummy gate electrodes **39d**. The dummy gate electrodes **39d** may be formed of the same material as the gate electrodes **39g** and the conductive layers **39c**.

The mold staircase region **MSA** may include interlayer insulating layers **33** and molded insulating layers **36b** repeatedly and alternately stacked. The molded insulating layers **36b** may be formed of silicon nitride.

In the stack structure **ST**, the dummy staircase region **ST_d** and the mold staircase region **MSA** may have a staircase shape.

When viewed from the plan view of FIG. **1** in the stack structure **ST**, the dummy staircase region **ST_d** and the mold staircase region **MSA** may be above and/or below a stack region including the first stack region **ST_1** and the second stack region **ST_2**.

When viewed from the plan view of FIG. **1** in the stack structure **ST**, the first and second stack regions **ST_1** and **ST_2** may include a central region having a planar upper surface and staircase regions **SA** on the left and right of the central region and having a staircase of which height is decreased.

The staircase region **SA** of the first stack region **ST_1** may be referred to as a gate staircase region **GSA**, and the staircase region **SA** of the second stack region **ST_2** may be referred to as a dummy staircase region **DSA**. For example, in the gate staircase region **GSA** of the first stack region **ST_1**, the gate electrodes **39g** may have gate pads **GP** arranged in a staircase shape and the second stack region may have gate pads **GP** arranged in a staircase shape. The conductive layers **39c** in the dummy staircase region **DSA** of the region **ST_2** may have a staircase shape.

The gate staircase region **GSA** of the first stack region **ST_1** may further include an insulator stack region **ST_i**. In the first stack region **ST_1**, the insulator stack region **ST_i** may include interlayer insulating layers **33** and molded insulating layers **36c** alternately stacked. The molded insulating layers **36c** may be formed of silicon nitride.

The semiconductor device **1** according to some example embodiments may further include a dam structure **65** that penetrates or extends through the gate staircase region **GSA** and surrounds the insulator stack region **ST_i** on the plate pattern **21**. The gate electrodes **39g** may be spaced apart from the molded insulating layers **36c** by the dam structure **65**.

In the plate pattern **21**, the gaps **21g1** and **21g2** may include a first gap **21g1** below the insulator stack region **ST_2b** of the second stack region **ST_2** and a second gap **21g2** below the insulator stack region **ST_i** of the gate staircase region **GSA** of the first stack region **ST_1**.

The first gap **21g1** may have a line shape or a bar shape extending in a first direction **X**. The insulator stack region **ST_2b** of the second stack region **ST_2** may cover the entire first gap **21g1** and may have a greater width than the first gap **21g1**.

The semiconductor device **1** according to some example embodiments may further include a first intermediate gap-fill pattern **24a** within the first gap **21g1**, a second intermediate gap-fill pattern **24b** within the second gap **21g2**, and an intermediate insulating layer **24c** on an external side surface of the plate pattern **21**. The first intermediate gap-fill pattern **24a**, the second intermediate gap-fill pattern **24b**, and the intermediate insulating layer **24c** may be formed of e.g., silicon oxide.

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The semiconductor device **1** according to some example embodiments may further include a capping insulating structure **93** on the stack structure **ST** and the intermediate insulating layer **24c**.

The vertical memory structures **50m** and the vertical dummy structures **50d** may penetrate through a central region of the first stack region **ST_1** that has a planar upper surface.

The first vertical support structures **50s1** may penetrate through the conductor stack region **ST_2a** in a central region of the first stacking region **ST_2** that has a planar upper surface.

The second vertical support structures **50s2** may penetrate through the staircase region **SA** of the first and second stack regions **ST_1** and **ST_2**.

The semiconductor device **1** according to an example embodiment may further include separation structures **63m**, **63s**, and **63d** penetrating through the stack structure **ST**.

The separation structures **63m**, **63s**, and **63d** may include main separation structures **63m** that intersect the stack structure **ST** in the first direction **X** and separate the stack structure **ST** in a second direction **Y**, auxiliary separation structures **63s** between the main separation structures **63m** and having a width smaller than a width of the stack structure **ST** in the first direction **X**, and dummy separation structures **63d** that penetrate through the staircase region **SA** the second stacking region **ST_2**.

The first direction **X** may be parallel to the upper surface of the semiconductor substrate **3**, and the second direction **Y** may be parallel to the upper surface of the semiconductor substrate **3** and perpendicular the first direction **X**.

The main separation structures **63m** may include a first main separation structure **63m1** and a second main separation structure **63m2** spaced apart from each other with the second stack region **ST_2** interposed therebetween. In the stack region **ST**, a plurality of the second stack regions **ST_2** may be spaced apart from each other in the second direction **Y**. Each of the plurality of second stack regions **ST2** may be in the form of a bar or a line extending in the first direction **X**.

The conductor stack region **ST_2a** of the second stack region **ST_2** may include a first stack portion (**ST_2aa** of FIG. 3A) between the first main separation structure **63m1** and the insulator stack region **ST_2b**, a second stack portion (**ST_2ab** of FIG. 3A) between the second main separation structure **63m2** and the insulator stacking region **ST_2b**, and a third stack portion (**ST_2ac** of FIG. 3A) connecting the first stack portion (**ST_2aa** of FIG. 3A) and the second stack portion (**ST_2ab** of FIG. 3A) to each other.

When viewed from plan view on a height level on which one of the conductive layers **39c** is disposed, the third stack portion (**ST_2ac** of FIG. 3A) may include convex portions in a direction toward the insulator stack region **ST_2b**.

When viewed from plan view on a height level on which one of the conductive layers **39c** is positioned, a first boundary **SB1** between the first stack portion (**ST_2aa** of FIG. 3A) and the insulator stack region **ST_2b** and a second boundary **SB2** between the second stack portion (**ST_2ab** of FIG. 3A) and the insulator stack region **ST_2b** may be in the form of a line extending in the first direction **X**. A third boundary **SB3** between the third stack portion (**ST_2ac** of FIG. 3A) and the insulator stack region **ST_2b** may be in the form of a curved surface or of curved surfaces.

The source contact plugs **70s** may be electrically connected to at least one of the conductive layers **39c** of the conductor stack region **ST_2a** and may be in contact with the plate pattern **21**. For example, the source contact plugs

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70s may penetrate through the upper layer **21c** and the first intermediate layer **21b** of the plate pattern **21** to be in contact with the lower layer **21a**. The source contact plugs **70s** may be in contact with the plate pattern **21** which may be a common source of a flash memory device.

At least one of the source contact plugs **70s** may penetrate between the insulator stack region **ST_2b** and the conductor stack region **ST_2a** and may contact the plate pattern **21**.

At least one of the source contact plugs **70s** may be in contact with an uppermost conductive layer, among the conductive layers **39c** of the conductor stack region **ST_2a**. At least one of the source contact plugs **70s** may be in contact with a plurality of conductive layers among the conductive layers **39c**, and a plurality of mold insulation layers among the molded insulating layers **36a**.

At least one of the conductive layers **39c** may have an upper surface and a side surface in contact with at least one of the source contact plugs **70s**. For example, at least one of the source contact plugs **70s** may be in contact with an upper surface and a side surface of an uppermost conductive layer, among the conductive layers **39c**.

As seen in FIG. 3A, the source contact plugs **70s** may include first side source contact plugs **70sa1** electrically connected to at least one of the conductive layers **39c** of the first stack portion **ST_2aa**, second side source contact plugs **70sa2** electrically connected to at least one of the conductive layers **39c** of the second stack portion **ST_2ab**, and third side source contact plugs **70sa3** electrically connected to at least one of the conductive layers **39c** of the third stack portion **ST_2ac**. The first source connection line **78s1** may overlap at least one of the first side source contact plugs **70sa1**, and may overlap at least one of the second side source contact plugs **70sa2**.

The semiconductor device **1** according to some example embodiments may further include external source contact plugs **70so** in contact with the plate pattern **21**, as seen in FIG. 4B. The external source contact plugs **70so** may penetrate through the mold staircase region **MSA** of the stack structure **ST** and the upper layer **21c** and the first intermediate layer **21b1** of the plate pattern **21**, and may be in contact with the lower layer **21a** of the plate pattern **21**.

The semiconductor device **1** according to some example embodiments may further include a first peripheral contact plug **72p1** penetrating through the insulator stack region **ST_2b** and the first intermediate gap-fill pattern **24a** of the second stack region **ST_2**. The first peripheral contact plug **72p1** may be electrically connected to the first peripheral pad **15p1** of the peripheral circuit structure **6**.

The semiconductor device **1** according to some example embodiments may further include a second peripheral contact plug **72p2** penetrating through the insulator stack region **ST_2b** and the second intermediate gap-fill pattern **24b** of the first stack region **ST_1**. The second peripheral contact plug **72p2** may be electrically connected to the second peripheral pad **15p2** of the peripheral circuit structure **6**.

The semiconductor device **1** according to some example embodiments may further include a third peripheral contact plug **72s** that may penetrate or extend through the intermediate insulating layer **24c** and may be electrically connected to the source pad **15s**. The third peripheral contact plug **72s** may be referred to as a source peripheral contact plug.

The first to third peripheral contact plugs **72p1**, **72p2**, and **72s** and the source contact plugs **70s** may have upper surfaces at substantially the same height level. Lower surfaces of the first to third peripheral contact plugs **72p1**, **72p2**, and **72s** may be arranged at a lower level than lower surfaces of the source contact plugs **70s**.

The first to third peripheral contact plugs **72p1**, **72p2**, and **72s**, the source contact plugs **70s**, and the external source contact plugs **70so** may be formed of the same material. For example, the first to third peripheral contact plugs **72p1**, **72p2**, and **72s**, the source contact plugs **70s**, and the external source contact plugs **70so** may include at least one of a metal nitride (for example, titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), or the like) and/or a metal (for example, tungsten (W), or the like).

The semiconductor device **1** according to an example embodiment may further include a bitline **78b** at a higher level than the stack structure **ST**.

The bitline **78b** may have a line shape extending in the second direction **Y**, and may intersect the first and second stack regions **ST_1** and **ST_2**.

The semiconductor device **1** according to an example embodiment may further include first bitline connection plug **75b1**, electrically connecting the vertical memory structure **50m** and the bitline **78b** to each other and between the vertical memory structure **50m** and the bitline **78b**, and a second bitline connection plug **75b2** electrically connecting the first peripheral contact plug **72p1** and the bitline **78b** to each other and between the first peripheral contact plug **72p1** and the bitline **78b**.

The first and second bitline connection plugs **75b1** and **75b2** may include a first plug (e.g., **75_1** of FIG. 5A) and a second plug (e.g., **75_2** of FIG. 5A) stacked in order.

The semiconductor device **1** according to some example embodiments may further include a first source connection line **78s1** and a source connection pattern **75s1** on the stack structure **ST**.

The first source connection line **78s1** may have a line shape extending in the second direction **Y**.

A plurality of first source connection line **78s1** may be disposed to be spaced apart from each other in the first direction **X**. Hereinafter, a single first source connection line **78s1** will mainly be described.

The source contact plugs **70s** include at least one first source contact plug **70s1**, which overlaps the first source connection line **78s1**, and second source contact plugs **70s2** which do not overlap the first source connection line **78s1**. At least one of the second source contact plugs **70s2** may overlap the bitline **78b**.

The source connection pattern **75s1** may electrically connect the first source connection line **78s1** and the at least one first source contact plug **70s1** to each other.

A width of the first source connection line **78s1** may be greater than a width of the bitline **78b**.

The width of the first source connection line **78s1** may be about three times to about ten times greater than the width of the bitline **78b**.

As illustrated in FIG. 4B, the semiconductor device **1** according to some example embodiments may further include a second source connection line **78s2**, extending in the first direction **X** on a higher level than the stack structure **S**, and a second source connection patterns **75s2** electrically connecting the second source connection line **78s** and the external source contact plugs **70so** between the source connection line **78s** and the external source contact plugs **70so**.

As illustrated in FIG. 4B, the semiconductor device **1** according to some embodiments may further include a first upper connection line **83s1** on a higher level than the first and second source connection lines **78s1** and **78s2**, a plug **81s1** electrically connecting the first upper connection line **83s1** and the first source connection line **78s1** and between the upper connection line **83s1** and the first source connec-

tion line **78s1**, and a plug **81s2** electrically connecting the first upper connection line **83s1** and the second source connection line **78s2** and between the first upper connection line **83s1** and the second source connection line **78s2**.

The source connection patterns **75s1** and **75s2**, the first and second source connection lines **78s1** and **78s2**, the plugs **81s1** and **81s2**, and the first upper connection line **83s1** may be defined as a first interconnection structure **84** for electrically connecting the source contact plugs **70s** and the external source contact plugs **70so** to each other.

As illustrated in FIG. 4B, the semiconductor device **1** according to some example embodiments may further include a second interconnection structure **88** electrically connecting the first interconnection structure **84** and the third peripheral contact plug **72s** to each other. The second interconnection structure **88** may include a second upper connection line **87**, a plug **85s1** electrically connecting the second upper connection line **87** and the first upper connection line **83s1** and between the second upper connection line **87** and the first upper connection line **83s1**, and a plurality of plugs **75s3**, **78s3**, **81s2**, **83s2**, and **85s2** electrically connecting the second upper connection line **87** and the third peripheral contact plugs **72s** and between the second upper connection line **87** and the third peripheral contact plugs **72s**.

Accordingly, an interconnection structure **84** and **88** electrically connecting the source contact plugs **70s**, the external source contact plugs **70so**, and the third peripheral contact plugs **72s** may include the first interconnection structure **84** and the second interconnection structure **88**.

In the semiconductor device **1** according to some example embodiments, the plate pattern **21**, which may be a common source of a flash memory device, may be electrically connected to the peripheral circuit of the peripheral circuit structure **6** through the source contact plugs **70s** in contact with the plate pattern **21**, at least one of the conductive layers **39c** in contact with the source contact plugs **70s**, and the external source contact plugs **70so**. Accordingly, the common source of the plate pattern **21** may be electrically connected to the peripheral circuit of the peripheral circuit structure **6** through the source contact plugs **70s** in contact with at least one of the conductive layers **39c** to address a noise issue occurring when the common source and the peripheral circuit are electrically connected to each other in the flash memory device. As a result, noise may be prevented from being generated by the common source in an operation, for example, in a read operation, of the flash memory device.

The semiconductor device **1** according to some example embodiments may further include gate contact plugs **67** on the gate pads **GP** of the gate electrodes **39g**, gate interconnections **78g** having a higher level than the gate contact plugs **67**, first gate connection plugs **75g1** between the gate interconnections **78g** and the gate contact plugs **67**, and second gate connection plugs **75g2** between the gate interconnections **78g** and the second peripheral contact plugs **72p2**.

The semiconductor device **1** may include a memory cell array region **1**, a buffer region **BA**, and a dummy region **DA**. The memory cell array region **MCA** may be a region in which the vertical memory structures **50m** are arranged. Alternatively, a portion of the first stack region **ST_1** of the stack structure **ST**, in which the vertical memory structures **50m** are arranged, may be defined as a memory cell array region **MCA**. The buffer region **BA** may be a region between the memory cell array region **MCA** and the staircase region **SA**, and the dummy region **DA** may be a region between the memory cell array region **MCA** and the second stack region

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ST_2. The dummy vertical structures **50d** may be disposed in the buffer region BA and the dummy region DA.

In the plan view of FIG. 1, a plurality of first source connection lines **78s1** may be disposed. The memory cell array region MCA may include first memory cell array regions MCA1 and second memory cell array regions MCA2. The first memory cell array regions MCA1 and the second memory cell array regions MCA2 may be disposed between the first source connection lines **78s1**. The second memory cell array regions MCA2 may be disposed between the second stack regions ST_2.

A width of each of the first memory cell array regions MCA1 in the second direction Y may be greater than a width of each of the second memory cell array regions MCA2 in the second direction Y.

The string separation pattern **46**, illustrated in FIG. 3B, may intersect and penetrate through a gate electrode, which may be an upper select gate electrode, among the plurality of upper gate electrodes (e.g., **39gU** of FIG. 5A). The dummy vertical structures **50d** may be in the buffer region BA and the dummy region DA, may penetrate through the string separation pattern **46** in the memory cell array region MCA, and may be below the first source connection line **78s1**.

Next, the vertical memory structure **50m** and the plate pattern **21** will mainly be described with reference to FIG. 5A. FIG. 5A is an enlarged cross-sectional view of region "D" of FIG. 4C.

Referring to FIG. 5A, a vertical memory structure **50m** may include an insulating core region **56m**, a pad pattern **58m** on the insulating core region **56m**, a dielectric structure **52m** on a side surface of the insulating core region **56m** and a side surface of the pad pattern **58m**, and a channel layer **54m** between the insulating core region **56m** and the dielectric structure **52m** and between the insulating core region **56m** and the pad pattern **58m**.

The pad pattern **58m** may be at a higher level than an uppermost gate electrode, among the gate electrodes **39g**.

The pad pattern **58m** may include doped silicon, for example, polysilicon having a first type conductivity (e.g., N-type conductivity). The channel layer **54m** may be in contact with the pad pattern **58m**. The channel layer **54m** may be formed of silicon. The dielectric structure **52m** may include a first dielectric layer **52a**, a data storage layer **52d**, and a second dielectric layer **52b**. The data storage layer **52d** may be interposed between the first dielectric layer **52a** and the second dielectric layer **52b**. The first dielectric layer **52a** may include silicon oxide and/or a high-k dielectric material. The data storage layer **52d** may include a material for storing data in a memory device, for example, silicon nitride for trapping charges. The second dielectric layer **52b** may be a tunnel dielectric layer in contact with the channel layer **54m**. The second dielectric layer **52b** may be silicon oxide or silicon oxide doped with impurities.

The gate electrodes **33g** may include a plurality of lower gate electrodes **33gL** and a plurality of upper gate electrodes **33gU** on the plurality of lower gate electrodes **33gL**.

The vertical memory structure **50m** may include a lower vertical portion **50L** that may penetrate or extend through the plurality of lower gate electrodes **33gL**, an upper vertical portion **50U** that may penetrate or extend through the plurality of upper gate electrodes **33gU**, and a width change portion **50V** between the vertical portion **50L** and the upper vertical portion **50U**.

The width-variable portion **50V** may be between an uppermost lower gate electrode among the plurality of lower

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gate electrodes **33gL**, and a lowermost upper gate electrode among the plurality of upper gate electrodes **33gU**.

The width change portion **50V** may have a side surface extending from and/or bending between a side surface of the lower vertical portion **50L** and a side surface of the upper vertical portion **50U**. The width change portion **50V** may have a side slope different from a side slope of the lower vertical portion **50L** and a side slope of the upper vertical portion **50U**. Accordingly, the width change portion **50V** may be referred to as a slope change portion or a bent portion.

The interlayer insulating layer **33**, which is between the uppermost lower gate electrode among the plurality of lower gate electrodes **33gL**, and the lowermost upper gate electrode among the plurality of upper gate electrodes **33gU**, may include a first layer **33_1** and a second layer **33_2** on the second layer **33_1**.

In the plate pattern **21**, the second intermediate layer **21b2** may penetrate or extend through the dielectric structure **52m** to be in contact with the channel layer **54m**. The dielectric structure **52m** may be divided into an upper dielectric structure **52mU** and a lower dielectric structure **52mL** by the second intermediate layer **21b2**.

The above-described molded insulating layers **36a** (of FIG. 4A) may include lower molded insulating layers **36aL**, having substantially the same height as the plurality of lower gate electrodes **33gL**, and upper molded insulating layers **36aU** having substantially the same height as the plurality of upper gate electrodes **33gU**. The conductive layers **39c** (of FIG. 4A) may include lower conductive layers **39cL**, having substantially the same height as the plurality of lower gate electrodes **33gL**, and upper conductive layers **39cU** having substantially the same height as the plurality of upper gate electrodes **33gU**.

In some embodiments, for example when the semiconductor device **1** is a flash memory device, at least one of the plurality of upper gate electrodes **39gU** may be an upper erase control gate electrode used in an erase operation of the flash memory device, and at least one of the gate electrodes **39gL** may be a lower erase control gate electrode used in an erase operation of the flash memory device. The number of erase control gates may be determined depending on the total number of stacked gate electrodes **39g**. Among the gate electrodes **39g** between the upper and lower erase control gate electrodes, a plurality of gate electrodes may be wordlines. Among the gate electrodes **39g**, a gate electrode between the lower erase gate electrode and the wordlines may be a ground select gate electrode, and a gate electrode between the upper erase gate electrode and the wordlines may be a string select gate electrode.

Next, the first and second vertical support structures **50s1** and **50s2**, the vertical dummy structures **50d**, and the plate pattern **21** will mainly be described with reference to FIG. 5B. FIG. 5B is an enlarged cross-sectional view of region "E" of FIG. 4C, and may represent a single second vertical support structure **50s2**. The first and second vertical support structures **50s1** and **50s2** and the vertical dummy structures **50d** may be formed of substantially the same layers and may have the same or similar cross-sectional structure.

Referring to FIG. 5B, the second vertical support structure **50s2** may include an insulating core region **56s**, a pad pattern **58s** on the insulating core region **56s**, a dielectric structure **52s** on a side surface of the insulating core region **56s** and a side surface of the pad pattern **58s**, and a channel layer **54s** between the insulating core region **56s** and the dielectric structure **52s** and between the insulating core region **56s** and the pad pattern **58s**.

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The insulating core region **56s**, the pad pattern **58s**, the dielectric structure **52s**, and the channel layer **54s** of the second vertical support structure **50s2** may correspond to the core region **56m**, the pad pattern **58m**, the dielectric structure **52m**, and the channel layer **54m** of the vertical memory structure **50m**, respectively.

In the plate pattern **21**, the first intermediate layer **21b1** may be in contact with the dielectric structure **52s** and may be spaced apart from the channel layer **54s**.

Hereinafter, among components of the semiconductor device **1** according to some example embodiments, some components which may be modified or replaced will mainly be described.

Next, a modified example of the cross-sectional structure of the source contact plugs **70s** will be described with reference to FIG. 6. FIG. 6 is a cross-sectional view illustrating regions taken along lines I-I' and II-II' of FIG. 3A.

Referring to FIG. 6, the source contact plug **70s** described of FIG. 4A may be replaced with a source contact plug **70s'** in contact with an upper surface, a side surface, and a lower surface of each of the conductive layers **39c**. The source contact plug **70s'** may be in contact with an upper surface a side surface, and a lower surface of each of the molded insulating layers **36a**.

Next, a modified example of the source contact plugs **70s** in plan view will be described with reference to FIG. 7. FIG. 7 is plan view illustrating a modified example of the source contact plugs **70s** in the plan view of FIG. 3A.

Referring to FIG. 7, in plan view, the source contact plugs **70s** of FIG. 3A may be replaced with source contact plugs **70s''** elongated in one direction. For example, the source contact plugs **70s''** may include source contact plugs **70sa3'** elongated in a first direction X and source contact plugs **70sa1'** and **70sa2'** elongated in a second direction Y.

Among the source contact plugs **70s''**, source contact plugs having a shape elongated in the first direction X may be in contact with a third stacked portion **ST_2ac**, and source contact plugs elongated in the second direction Y may be in contact with first and second stacked portions **ST_2aa** and **ST_2ab**.

Next, a modified example of the source contact plugs **70s** will be described with reference to FIG. 8. FIG. 8 may illustrate a modified example of the source contact plugs **70s** and the source connection pattern **75s1** in the cross-sectional view of FIG. 4A illustrating regions taken along lines I-I' and II-II' of FIG. 3A.

Referring to FIG. 8, the source contact plugs **70s** of FIG. 4A may be replaced with source contact plugs **170s** that are spaced apart from the conductor stack region **ST_2a** and that penetrate or extend through the insulator stack region **ST_2b** to be in contact with the plate pattern **21**. The semiconductor device **1** according to some example embodiments may further include connection structures **73a** and **73b** electrically connecting at least one of the conductive layers **39c** and each of the source contact plugs **170s**. The connection structures **73a** and **73b** may include first patterns **73a**, arranged to be in contact with at least an uppermost conductive layer among the conductive layers **39c**, and second patterns **73b** electrically connecting the first patterns **73a** and the source contact plugs **170s** to each other.

The source contact plugs **170s** may include a first source contact plug **170s1** overlapping the first source connection line **78s1**, and second source contact plugs **170s2** that do not overlap the first source connection line **78s1**.

The source connection pattern **75s1** described in FIG. 4A may be replaced with a source connection pattern **75s1'** electrically connecting the second pattern **73b** and the first

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source connection line **78s1** between the first source connection line **78s1** and the second pattern **73b** electrically connected to the first source contact plug **170s**.

Next, a modified example of the dummy separation structures **63d** and the third boundary **SB3** between the third stack portion **ST_2ac** and the insulator stack region **ST_2b**, described with reference to FIG. 3A, in plan view will be described with reference to FIG. 9.

Referring to FIG. 9, in the plan view, the third boundary **SB3** between the third stack portion **ST_2ac** and the insulator stack region **ST_2b** of FIG. 3A may be replaced with a third boundary **SB3'** having a linear shape and extending in the second direction. The dummy separation structures **63d** of FIG. 3A may be replaced with dummy separation structures **63d'** including a first line portion, extending in the second direction Y between the first and second main separation structures **63m1** and **63m2**, and second line portions disposed in a direction away from the insulator stack region **ST_2b** from the first line portion.

Next, a modified example of the second vertical support structures **50s2** of FIG. 4C will be described with reference to FIG. 10. FIG. 10 may illustrate a modified example of the second vertical support structures **50s2** in the cross-sectional view of FIG. 4C illustrating regions taken along line IV-IV' of FIG. 3B and line V-V of FIG. 1.

Referring to FIG. 10, the second vertical support structures **50s2** be modified into second vertical support structures **150s2** having a structure different from the structure of the first vertical support structures **50s1** and the structure of the vertical dummy structures **50d**. For example, the first vertical support structures **50s1** and the vertical dummy structures **50d** may be formed to include the insulating core region **56s**, the pad pattern **58s**, and the dielectric structure **52s**, and the channel layer **54s** as in FIG. 5B, and the second vertical support structures **50s2** may not include the pad pattern **58s**, the dielectric structure **52s**, and the channel layer **54s**. The second vertical support structures **50s2** may be formed of pillars formed of silicon oxide.

Next, a modified example of the source contact plugs **70s** will be described with reference to FIGS. 11 to 13. FIG. 11 is an enlarged plan view of region "A" of FIG. 1, FIG. 12 is an enlarged plan view of region "Ba" of FIG. 11, and FIG. 13 is a cross-sectional view illustrating regions taken along line Ia-Ia' and IIa-IIa' of FIG. 12.

Referring to FIGS. 11 to 13, in FIGS. 2, 3A and 4A, the first vertical support structures **50s1** penetrating through the conductor stack region **ST_2a** may be omitted, and the source contact plugs **70s** may be replaced with source contact plugs **270s** that penetrate or extend through the conductor stack region **ST_2a** and are arranged to be in contact with the plate pattern **21**. The source contact plugs **270s** may be spaced apart from the insulator stack region **ST_2b**.

The source contact plugs **270s** may be in contact with the conductive layers **39c**.

The source contact plugs **270s** may be in contact with a side surface, an upper surface, and a lower surface of each of the conductive layers **39c**. The source contact plugs **270s** may include a first source contact plug **270s1** overlapping the first source connection line **78s1**, and second source contact plugs **270s2** that are not overlapping the first source connection line **78s1**.

The source contact plugs **270s** may include first side source contact plugs penetrating or extending through the first stack portion **ST_2aa**, second side source contact plugs penetrating or extending through the second stack portion **ST_2ab**, and the third side source contact plugs penetrating

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or extending through the stack portion **ST_{2ac}**. The first source connection line **78s1** may overlap at least one of the first side source contact plugs, and may overlap at least one of the second side source contact plugs.

Next, a modified example of the source contact plugs **270s** of FIG. 13 will be described with reference to FIG. 14. FIG. 14 is a cross-sectional view illustrating regions taken along lines Ia-Ia' and IIa-IIa' of FIG. 12.

Referring to FIG. 14, the source contact plugs **270s** of FIG. 13 may be modified into source contact plugs **270s'** including a width change portion **270sv** having a width changed on a higher level than the stack structure **ST**. For example, the source contact plugs **270s'** may include a first portion **270sa**, a second portion **270sb** having a greater width than the first portion **270sa** on the first portion **270sa**, and the width change portion **270sv** formed due to a difference in width between the first portion **270sa** and the second portion **270sb**. The second portion **270sb** may be at a higher level than the stack structure **ST**.

The above-described first peripheral contact plug **72p1** may be modified into a first peripheral contact plug **72p1'** having the same shape as the source contact plugs **270s'** on a higher level than the stack structure **ST**.

Next, a modified example of the first intermediate layer **21b1** of the plate pattern **21** will be described with reference to FIG. 15. FIG. 15 is a cross-sectional view illustrating regions taken along lines Ia-Ia' and IIa-IIa' of FIG. 12.

Referring to FIG. 15, the first intermediate layer (**21b1** of FIGS. 4A, 6, 8, and 13) of the above-described plate pattern **21** may be in contact with the main separation structure **63m**, and may be modified into a first intermediate layer **21b1'** including doped polysilicon.

Next, a modified example of the third boundary **SB3'** between the third stack portion **ST_{2ac}** and the insulator stack region **ST_{2b}** in FIG. 12 and the dummy separation structures **63d** in FIG. 11 in plan view will be described with reference to FIG. 16.

Referring to FIG. 16, in the plan view, the third boundary **SB3'** between the third stack portion **ST_{2ac}** and the insulator stack region **ST_{2b}** of FIG. 12 may be modified into a third boundary **SB3'** in the form of a line extending in the second direction **Y** as in FIG. 9. The dummy separation structures **63d** of FIG. 11 may be modified into dummy separation structures **63d'** including a first line portion, extending in the second direction **Y** between the first and second main separation structures **63m1** and **63m2** as in FIG. 9, and dummy separation structures **63d''** arranged in a direction away from the insulator stacking region **ST_{2b}** from the first line portion.

Next, a modified example of the first and second interconnection structures **84** and **88** described with reference to FIG. 4B will be described with reference to FIG. 17. FIG. 17 is a cross-sectional view illustrating regions taken along line of FIG. 1.

Referring to FIG. 17, the first upper connecting line **83s1** described with reference to FIG. 4B may be modified into an upper connection line **83s** extending to an upper portion of the third peripheral contact plug **72s**. The upper connection line **83s** may be electrically connected to the third peripheral contact plug **72s** through a plurality of plugs **75s3**, **78s3**, and **81s2** between the upper connection line **83s** and the third peripheral contact plug **72s**.

Next, a modified example of the first and second interconnection structures **84** and **88** described with reference to FIG. 4B will be described with reference to FIG. 18. FIG. 18 is a cross-sectional view illustrating regions taken along line of FIG. 1.

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Referring to FIG. 18, the first source connection line **78s1** of FIG. 4B may be modified into a source connection line **78s1'** extending to upper portions of the external source contact plugs **70s0** and to an upper portion of the third peripheral contact plug **72s**. The source connecting line **78s1'** may be electrically connected to the external source contact plugs **70s0** through the plugs **75s2** between the source connecting line **78s1'** and the external source contact plugs **70s0**, and may be electrically connected to the third peripheral contact plugs **72s** through a plug **75s3** that is between the third peripheral contact plugs **72s** and the source connecting line **78s1'**.

Next, a modified example of the external source contact plugs **70s0** described with reference to FIG. 4B will be described with reference to FIG. 19. FIG. 19 is a cross-sectional view illustrating a modified example of the external source contact plugs **70s0** of FIG. 4B.

Referring to FIG. 19, in FIG. 4B, the external source contact plugs **70s0** that penetrate or extend through the mold staircase region **MSA** of the stack structure **ST** to be in contact with the plate pattern **21** may be modified into external source contact plugs **70s0'** spaced apart from the stack structure **ST** and in contact with the plate pattern **21**.

An example of a method of forming a semiconductor device according to some example embodiments will be described with reference to FIG. 20. FIG. 20 is a process flowchart illustrating an example of a method of forming a semiconductor device according to some example embodiments.

Referring to FIG. 20, in operation **S10**, a peripheral circuit structure (e.g., **6** of FIGS. 4A to 4C) may be formed on a semiconductor substrate (e.g., **3** of FIGS. 4A to 4C). In operation **S20**, a plate pattern (e.g., **21** of FIGS. 4A and 4B), having gaps (e.g., **21g1** and **21g2** of FIGS. 4A and 4B) spaced apart from each other, may be formed on the peripheral circuit structure. Intermediate gap-fill patterns (e.g., **24a** and **24b** of FIGS. 4A to 4C) within the gaps of the plate pattern **21** and an intermediate insulating layer (e.g., **24c** of FIGS. 4A to 4C) on the external side surface of the plate pattern **21** may be simultaneously formed or formed in a common operation.

In operation **S30**, a preliminary stack structure including interlayer insulating layers **33** and molded insulating layers alternately stacked on the plate pattern **21** may be formed. In operation **S40**, vertical memory structures penetrating through the preliminary stack structure may be formed. While forming the vertical memory structures, vertical support structures (e.g., **50s1** of FIG. 4A, **50s2** of FIG. 4C, and **50s2** of FIG. 5B) and vertical dummy structures (e.g., **50d** of FIG. 4C) may be simultaneously formed or formed in a common operation.

In operation **S40**, separation trenches (e.g., **60** of FIG. 4A) penetrating through the preliminary stack structure may be formed.

In some embodiments, a portion of the plate pattern **21** may be replaced with a polysilicon material layer to form a second intermediate layer **21b2** as illustrated in FIGS. 4C and 5A. In some embodiments, the second intermediate layer (e.g., **21b2** of FIGS. 4C and 5A) may penetrate through the dielectric structure (e.g., **52m** of FIGS. 4C and 5A) of the vertical memory structures (e.g., **50m** of FIGS. 4C and 5A) to be in contact with the channel layer (e.g., **54m** of FIGS. 4C and 5A).

In operation **S60**, some of the molded insulating layers may be replaced with gate layers to form a stack structure (e.g., **ST** of FIGS. 1 to 5B) including a first stack region

(e.g., ST_1 of FIGS. 1 to 5B) and a second stack region (e.g., ST_2 of FIGS. 1 to 5B) ST of 5b).

The gate layers may include gate electrodes 39g and 39d and the conductive layers 39c as described with reference to FIGS. 1 to 5B. The gate layers may further include the dielectric layer 41d as described with reference to FIGS. 1 to 5B. Among the molded insulating layers, remaining molded insulating layers may be the molded insulating layers 36a, 36b, and 36c as described with reference to FIGS. 4A to 4C.

In operation S70, separation structures (e.g., 60m, 60s, and 60d of FIGS. 1 to 4C) may be formed to fill the isolation trenches (e.g., 60 of FIG. 4A).

In operation S80, an interconnection structure may be formed. The interconnection structure may be conductive patterns for electrically connecting the plate pattern 21, the vertical memory structures 50m, and the gate electrodes 39g to the peripheral circuit 12 of the peripheral circuit structure 6. Such an interconnection structure may include various components as described with reference to FIGS. 1 to 19, for example, the source contact plugs (e.g., 70 of FIG. 4A), the external source contact plugs (e.g., 70so of FIG. 4B), the gate contact plug (e.g., 67 of FIG. 4C), the first to third peripheral contact plugs (e.g., 72p1, 72p2, 72s of FIGS. 4A to 4C), the bitline (e.g., 78b of FIGS. 4A and 4C), the gate connection line (e.g., 78g of FIG. 4C), and the interconnection structure (e.g., 84 and 88 of FIGS. 1 to 4C).

Next, a data storage system including a semiconductor device according to some example embodiments will be described with reference to each of FIGS. 21, 22, and 23.

FIG. 21 is a schematic diagram illustrating a data storage system including a semiconductor device according to some example embodiments.

Referring to FIG. 21, a data storage system 1000 in some example embodiments may include a semiconductor device 1100 and a controller 1200 electrically connected to the semiconductor device 1100 to control the semiconductor device 1100. The data storage system 1000 may be implemented by a storage device including the semiconductor device 1100 or an electronic device including a storage device. For example, the data storage system 1000 may be implemented by a solid-state drive device (SSD), a universal serial bus (USB), a computing system, a medical device, or a communication device, including the semiconductor device 1100.

In some example embodiments, the data storage system 1000 may be implemented by an electronic system storing data.

The semiconductor device 1100 may be implemented by a semiconductor device described in the above-described example embodiments with reference to FIGS. 1 to 19. The semiconductor device 1100 may include a first structure 1100F and a second structure 1100S on the first structure 1100F.

The first structure 1100F may be a peripheral circuit structure including a decoder circuit 1110, a page buffer 1120, and a logic circuit 1130. For example, the first structure 1100F may include the peripheral circuit structure (6 of FIGS. 4A to 4C) including the above-described peripheral circuit (12 of FIGS. 4A to 4C). The peripheral circuit (12 of FIGS. 4A to 4C) may be a transistor constituting a peripheral circuit structure including the decoder circuit 1110, the page buffer 1120, and the logic circuit 1130.

The second structure 1100S may be a memory structure including a bitline BL, a common source line CSL, wordlines WL, first and second gate upper lines UL1 and UL2,

first and second gate lower lines LL1 and LL2, and memory cell strings CSTR between the bitline BL and the common source CSL.

In the second structure 1100S, each of the memory cell strings CSTR may include lower transistors LT1 and LT2 adjacent to the common source line CSL, upper transistors UT1 and UT2 adjacent to the bitline BL, and a plurality of memory cell transistors MCT between the lower transistors LT1 and LT2 and the upper transistors UT1 and UT2. The number of the lower transistors LT1 and LT2 and the number of the upper transistors UT1 and UT2 may vary in example embodiments.

In some example embodiments, the upper transistors UT1 and UT2 may include a string select transistor, and the lower transistors LT1 and LT2 may include a ground select transistor. The lower gate lines LL1 and LL2 may be gate electrodes of lower transistors LT1 and LT2, respectively. The wordlines WL may be gate electrodes of memory cell transistors MCT, and the gate upper lines UL1 and UL2 may be gate electrodes of the upper transistors UT1 and UT2, respectively.

The above-described gate horizontal layers (39G of FIGS. 4C and 5A) may constitute the lower gate lines LL1 and LL2, the wordlines WL, and the upper gate lines UL1 and UL2.

In some example embodiments, the lower transistors LT1 and LT2 may include a lower erase control transistor LT1 and a ground select transistor LT2 connected to each other in series. The upper transistors UT1 and UT2 may include a string select transistor UT1 and an upper erase control transistor UT2 connected to each other in series. At least one of the lower erase control transistor LT1 and the upper erase control transistor UT1 may be used for an erase operation to erase data, stored in the memory cell transistors MCT, using a gate-induced drain leakage (GIDL) phenomenon.

The common source line CSL, the first and second gate lower lines LL1 and LL2, word lines WL, and the first and second gate upper lines UL1 and UL2 may be electrically connected to the decoder circuit 1110 through first connection lines 1115 extending from the first structure 1100F to the second structure 1100S.

At least a portion of the above-described plate pattern (21 of FIGS. 4A to 4C) may be the common source line CSL. The bitlines BL may be electrically connected to the page buffer 1120 through second connection lines 1125 extending from the first structure 1100F to the second structure 1100S. The bitlines BL may be the above-described bitlines 78b.

In the first structure 1100F, the decoder circuit 1110 and the page buffer 1120 may perform a control operation on at least one select memory cell transistor of the plurality of memory cell transistors MCT. The decoder circuit 1110 and the page buffer 1120 may be controlled by a logic circuit 1130.

The semiconductor device 1000 may further include an input/output pad 1101.

The semiconductor device 1100 may communicate with the controller 1200 through the input and output pad 1101 electrically connected to the logic circuit 1130. The input/output pad 1101 may be electrically connected to the logic circuit 1130 through an input/output connection line 1135 extending from the first structure 1100F to the second structure 1100S. Accordingly, the controller 1200 may be electrically connected to the semiconductor device 1000 through the input/output pad 1101 and may control the semiconductor device 1000.

The controller 1200 may include a processor 1210, a NAND controller 1220, and a host interface 1230. Accord-

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ing to some example embodiments, the data storage system **1000** may include a plurality of semiconductor devices **1100**. In this case, the controller **1200** may control the plurality of semiconductor devices **1000**.

The processor **1210** may control overall operation of the data storage system **1000** including the controller **1200**. The processor **1210** may operate according to a predetermined firmware and may control the NAND controller **1220** to access the semiconductor device **1100**. The NAND controller **1220** may include a NAND interface **1221** for processing communication with the semiconductor device **1100**. A control command for controlling the semiconductor device **1100**, data to be written in the memory cell transistors MCT of the semiconductor device **1100**, and/or data to be read from the memory cell transistors MCT of the semiconductor device **1100** may be transmitted through the NAND interface **1221**. The host interface **1230** may provide a communication function between the data storage system **1000** and an external host. When a control command is received from an external host through the host interface **1230**, the processor **1210** may control the semiconductor device **1100** in response to the control command.

FIG. **22** is a schematic perspective view illustrating a data storage system including a semiconductor device according to some example embodiments.

Referring to FIG. **22**, a data storage system **2000** in some example embodiments may include a main substrate **2001**, a controller **2002** mounted on the main substrate **2001**, one or more semiconductor packages **2003**, and a DRAM **2004**. The semiconductor package **2003** and the DRAM **2004** may be connected to the controller **2002** by interconnection patterns **2005** formed on the main substrate **2001**.

The main substrate **2001** may include a connector **2006** including a plurality of pins coupled to an external host. The number and arrangement of the plurality of pins in the connector **2006** may be varied according to a communication interface between the data storage system **2000** and the external host. In some example embodiments, the data storage system **2000** may communicate with an external host according to one of interfaces such as a universal serial bus (USB), peripheral component interconnect express (PCI-Express), serial advanced technology attachment (SATA), and M-PHY for universal flash storage (UFS). In some example embodiments, the data storage system **2000** may operate by power supplied from an external host through the connector **2006**. The data storage system **2000** may further include a power management integrated circuit (PMIC) for distributing power supplied from the external host to the controller **2002** and the semiconductor package **2003**.

The controller **2002** may write data in the semiconductor package **2003** or may read data from the semiconductor package **2003** and may improve an operation speed of the data storage system **2000**.

The DRAM **2004** may be implemented by a buffer memory for reducing a difference in speed between the semiconductor package **2003**, a data storage space, and an external host. The DRAM **2004** included in the data storage system **2000** may also operate as a type of cache memory and may provide a space for temporarily storing data in a control operation performed on the semiconductor package **2003**. When the DRAM **2004** is included in the data storage system **2000**, the controller **2002** may further include a DRAM controller for controlling the DRAM **2004** in addition to the NAND controller for controlling the semiconductor package **2003**.

The semiconductor package **2003** may include first and second semiconductor packages **2003a** and **2003b** spaced

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apart from each other. Each of the first and second semiconductor packages **2003a** and **2003b** may be configured as a semiconductor package including a plurality of semiconductor chips **2200**. Each of the semiconductor chips **2200** may include a semiconductor device, such as the example embodiments described above with reference to FIGS. **1** to **19**.

Each of the first and second semiconductor packages **2003a** and **2003b** may include a package substrate **2100**, semiconductor chips **2200** on the package substrate **2100**, adhesive layers **2300** on a lower surface of each of the semiconductor chips **2200**, a connection structure **2400** electrically connecting the semiconductor chips **2200** to the package substrate **2100**, and a molding layer **2500** covering the semiconductor chips **2200** and the connection structure **2400** on the package substrate **2100**.

The package substrate **2100** may be implemented by a printed circuit board including package upper pads **2130**. Each of the semiconductor chips **2200** may include an input/output pad **2210**.

In some example embodiments, the connection structure **2400** may be configured as a bonding wire electrically connecting the input and output pad **2210** to the package upper pads **2130**. Accordingly, in each of the first and second semiconductor packages **2003a** and **2003b**, the semiconductor chips **2200** may be electrically connected to each other by a bonding wire method and may be electrically connected to the package upper pads **2130** of the package substrate **2100**. In some example embodiments, in each of the first and second semiconductor packages **2003a** and **2003b**, the semiconductor chips **2200** may be electrically connected to each other by a connection structure including a through-silicon via (TSV), instead of the connection structure **2400** of a bonding wire method.

In some example embodiments, the controller **2002** and the semiconductor chips **2200** may be included in a single package. For example, the controller **2002** and the semiconductor chips **2200** may be mounted on an interposer substrate different from the main substrate **2001**, and the controller **2002** may be connected to the semiconductor chips **2200** through an interconnection formed on the interposer substrate.

FIG. **23** is a schematic cross-sectional view illustrating a data storage system including a semiconductor device according to some example embodiments. FIG. **23** illustrates an example embodiment of the semiconductor package **2003** illustrated in FIG. **22**, and illustrates a cross-sectional region of the semiconductor package **2003** illustrated in FIG. **22** taken along line VI-VI'.

Referring to FIG. **23**, in the semiconductor package **2003**, the package substrate **2100** may be configured as a printed circuit board. The package substrate **2100** may include a package substrate body portion **2120**, package upper pads **2130** on an upper surface of the package substrate body portion **2120**, lower pads **2125** on a lower surface of the package substrate body portion **2120** or exposed through the lower surface, and internal interconnections **2135** electrically connecting the package upper pads **2130** to the lower pads **2125** in the package substrate body portion **2120**. The package upper pads **2130** may be electrically connected to the connection structures **2400**. The lower pads **2125** may be connected to interconnection patterns **2005** of a main substrate **2001** of the data storage system **2000** through conductive connection portions **2800** as illustrated in FIG. **22**.

Each of the semiconductor chips **2200** may include a semiconductor substrate **3010** and a first structure **3100** and a second structure **3200** stacked in order on the semicon-

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ductor substrate **3010**. The first structure **3100** may include a peripheral circuit region including peripheral interconnections **3110**. The second structure **3200** may include a common source line **3205**, a gate stack structure **3210** on the common source line **3205**, memory channel structures **3220** and separation structures **3230** penetrating through the gate stack structure **3210**, bitlines **3240** electrically connected to the memory channel structures **3220**, and gate connection lines **93b** electrically connected to wordlines (WL of FIG. **21**) of the gate stack structure **3210**. The first structure **3100** may include the first structure **1100F** illustrated in FIG. **21**, and the second structure **3200** may include the second structure **1100S** illustrated in FIG. **21**. For example, of FIG. **23**, a partially enlarged region denoted by reference numeral **1** may represent a cross-sectional structure of FIG. **4A**. Accordingly, each of the semiconductor chips **2200** may include a semiconductor device **1** according to one of the embodiments described above with reference to FIGS. **1** to **19**.

Each of the semiconductor chips **2200** may include a through-interconnection **3245** electrically connected to the peripheral interconnections **3110** of the first structure **3100** and extending into the second structure **3200**. The through-interconnection **3245** may penetrate through the gate stack structure **3210** and may be on an external side of the gate stack structure **3210**.

Each of the semiconductor chips **2200** may further include an input/output connection line **3265**, electrically connected to the peripheral interconnections **3110** of the first structure **3100** and extending inwardly of the second structure **3200**, and an input/output pad **2210** electrically connected to the input/output connection line **3265**.

As described above, according to some example embodiments, gate electrodes may be vertically stacked to improve integration density of a semiconductor device. Among conductive layers at the same height level as the vertically stacked gate electrodes, at least one conductive layer may be electrically connected to a plate pattern, which may be a common source, through source contact plugs. Therefore, the plate pattern used as a common source of a flash memory device may be electrically connected to a peripheral circuit through at least one of the source contact plugs and the conductive layers, so that noise may be prevented from being generated by the common source during an operation, for example, during a read operation, of the flash memory device. As a result, a semiconductor device having improved integration density and improved electrical performance may be provided.

While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concepts as defined by the appended claims.

What is claimed is:

1. A semiconductor device comprising:
 - a semiconductor substrate;
 - a peripheral circuit structure on the semiconductor substrate;
 - a plate pattern having a gap and on the peripheral circuit structure;
 - a stack structure on the plate pattern and including a first stack region and a second stack region spaced apart from the first stack region, wherein the first stack region includes gate electrodes stacked in a vertical direction that is perpendicular to an upper surface of the semiconductor substrate, wherein the second stack region includes a conductor stack region including conductive

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layers stacked in the vertical direction, and wherein the second stack region includes an insulator stack region including molded insulating layers at substantially the same height level as the conductive layers;

- a vertical memory structure that extends through the first stack region; and
- source contact plugs electrically connected to at least one of the conductive layers of the conductor stack region and in contact with the plate pattern.

2. The semiconductor device of claim 1, further comprising:

- a bitline on the stack structure; and
- a first peripheral contact plug that extends through at least the insulator stack region and is electrically connected to a first peripheral pad of the peripheral circuit structure;

- a first bitline connection plug that electrically connects the vertical memory structure and the bitline to each other; and

- a second bitline connection plug that electrically connects the first peripheral contact plug and the bitline to each other, wherein:

the gap has a line shape or a bar shape extending in a first direction that is parallel to the upper surface of the semiconductor substrate; and

the bitline has a line shape or a bar shape extending in a second direction that is parallel to the upper surface of the semiconductor substrate and is perpendicular to the first direction.

3. The semiconductor device of claim 2, further comprising:

- a first source connection line on the stack structure; and
- a source connection pattern,

wherein the source contact plugs include at least one first source contact plug that overlap the first source connection line, and second source contact plugs that do not overlap the first source connection line;

wherein at least one of the second source contact plugs overlaps the bitline;

wherein the source connection pattern electrically connects the first source connection line with the first source contact plug;

wherein the first source connection line has a line shape extending in the second direction; and

wherein a width of the first source connection line is greater than a width of the bitline.

4. The semiconductor device of claim 3, wherein:

the width of the first source connection line is about three times to ten times greater than the width of the bitline.

5. The semiconductor device of claim 3, wherein:

at least one of the source contact plugs is between the insulator stack region and the conductor stack region; and

the source connection pattern electrically connects the first source contact plug with the first source connection line.

6. The semiconductor device of claim 3, further comprising:

first patterns contacting at least an uppermost conductive layer among the conductive layers of the conductor stack region; and

second patterns electrically connecting the first patterns and the source contact plugs to each other,

wherein the source contact plugs are electrically connected to at least one of the conductive layers through the first and second patterns; and

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wherein the source connection pattern is in contact with and electrically connected to a second pattern, electrically connected to the first source contact plug, among the second patterns.

7. The semiconductor device of claim 3, further comprising:

external source contact plugs spaced apart from the first and second stack regions and contacting the plate pattern; and

a second source connection line electrically connecting the external source contact plugs to each other on the external source contact plugs,

wherein the source contact plugs and the external source contact plugs have upper surfaces at an equal distance from the upper surface of the semiconductor substrate; and

wherein the second source connection line and the first source connection line are an equal distance from the upper surface of the semiconductor substrate.

8. The semiconductor device of claim 7, further comprising:

a source peripheral contact plug electrically connected to a source pad of the peripheral circuit structure and having an upper surface having a level farther from the upper surface of the semiconductor substrate than a level of an uppermost gate electrode among the gate electrodes is from the upper surface of the semiconductor substrate;

a first upper interconnection structure electrically connecting the first source connection line with the second source connection line; and

a second upper interconnection structure electrically connecting the first upper interconnection structure with the source peripheral contact plug.

9. The semiconductor device of claim 1, further comprising:

separation structures that extend through the stack structure,

wherein, in a plan view, the separation structures include a first main separation structure and a second main separation structure opposing each other with the gap interposed therebetween;

wherein the second stack region is between the first main separation structure and the second main separation structure; and

wherein the conductor stack region of the second stack region comprises:

a first stack portion between the first main separation structure and the insulator stack region;

a second stack portion between the second main separation structure and the insulator stack region; and

a third stack portion connecting the first stack portion with the second stack portion.

10. The semiconductor device of claim 9, wherein: the gap has a line shape or a bar shape that extends in a first direction that is parallel to the upper surface of the semiconductor substrate; and

when viewed in a plan view from a height level on which one of the conductive layers is arranged, the third stack portion includes portions convex in a direction toward the insulator stack region.

11. The semiconductor device of claim 9, wherein: when viewed in a plan view from a height level of one of the conductive layers, a first boundary between the first stack portion and the insulator stack region and a second boundary between the second stack portion and the insulator stack region are in a form of a line

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extending in a first direction that is parallel to the upper surface of the semiconductor substrate; and

a third boundary between the third stack portion and the insulator stack region extends in a second direction that is parallel to the upper surface of the semiconductor substrate and perpendicular to the first direction.

12. The semiconductor device of claim 9, further comprising:

a source connection line on the stack structure,

wherein the source contact plugs include first side source contact plugs that extend through the first stack portion, second side source contact plugs that extend through the second stack portion, and third side source contact plugs that extend through the third stack portion; and

wherein the source connection line overlaps at least one of the first side source contact plugs and overlaps at least one of the second side source contact plugs.

13. The semiconductor device of claim 1, wherein:

a first of the source contact plugs is elongated in a first direction that is parallel to the upper surface of the semiconductor substrate and a second of the source contact plugs is elongated in a second direction that is parallel to the upper surface of the semiconductor substrate and perpendicular to the first direction.

14. The semiconductor device of claim 1, wherein:

at least one of the source contact plugs is in contact with two or more of the conductive layers.

15. The semiconductor device of claim 1, wherein:

at least one of the conductive layers has an upper surface and a side surface in contact with at least one of the source contact plugs.

16. The semiconductor device of claim 15, wherein:

the at least one of the conductive layers has a lower surface in contact with the at least one of the source contact plugs.

17. The semiconductor device of claim 1, wherein:

each of the source contact plugs includes a slope change portion at which a slope of a side wall thereof is changed, the slope change portion at a level higher than a level of an uppermost gate electrode among the gate electrodes.

18. The semiconductor device of claim 1, further comprising:

vertical support structures penetrating through the conductor stack region.

19. A semiconductor device comprising:

a semiconductor substrate;

a peripheral circuit structure on the semiconductor substrate;

a plate pattern on the peripheral circuit structure and having gaps therein that are spaced apart from each other;

a stack structure on the plate pattern and including first stack regions spaced apart from each other and second stack regions spaced apart from each other, wherein each of the first stack regions includes gate electrodes stacked in a vertical direction that is perpendicular to an upper surface of the semiconductor substrate, wherein each of the second stack regions includes a conductor stack region including conductive layers stacked in the vertical direction, and wherein each of the second stack regions includes an insulator stack region including molded insulating layers at substantially the same height level as the conductive layers;

vertical memory structures that extend through the first stack regions;

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source contact plugs that extend through the second stack regions;
 separation structures that extend through the stack structure;
 source connection lines having a higher level than the stack structure; and
 bitlines having a higher level than the stack structure, wherein:
 when viewed in a plan view, each of the gaps has a line shape or a bar shape that extend in a first direction that is parallel to the upper surface of the semiconductor substrate;
 when viewed in the plan view the first and second stack regions are spaced apart from each other in a second direction that is parallel to the upper surface of the semiconductor substrate and perpendicular to the first direction;
 the insulator stack regions of the second stack regions overlap the gaps;
 in each of the second stack regions, at least one of the conductive layers of the second stack region is electrically connected to the source contact plugs that extend through the second stack region;
 each of the source connection lines has a line shape that extends in the second direction;
 each of the bitlines has a line shape that extends in the second direction; and
 the source contact plugs include first source contact plugs that overlap the source connection lines, and second source contact plugs that overlap the bitlines.

20. A data storage system comprising:
 a semiconductor device including an input/output pattern; and
 a controller electrically connected to the semiconductor device through the input/output pattern and configured to control the semiconductor device, wherein:
 the semiconductor device comprises:
 a semiconductor substrate;
 a peripheral circuit structure on the semiconductor substrate;
 a plate pattern on the peripheral circuit structure and having gaps spaced apart from each other;
 a stack structure on the plate pattern and including first stack regions spaced apart from each other and

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second stack regions spaced apart from each other, wherein each of the first stack regions includes gate electrodes stacked in a vertical direction that is perpendicular to an upper surface of the semiconductor substrate, wherein each of the second stack regions includes a conductor stack region including conductive layers stacked in the vertical direction, and wherein each of the second stack regions includes an insulator stack region including molded insulating layers disposed at substantially the same height level as the conductive layers;
 vertical memory structures that extend through the first stack regions;
 source contact plugs that extend through the second stack regions;
 separation structures that extend through the stack structure;
 source connection lines having a higher level than the stack structure; and
 bitlines having a higher level than the stack structure, wherein:
 when viewed in a plan view, each of the gaps has a line shape or a bar shape that extend in a first direction that is parallel to the upper surface of the semiconductor substrate;
 when viewed in the plan view, the first and second stack regions are spaced apart from each other in a second direction that is parallel to the upper surface of the semiconductor substrate and perpendicular to the first direction;
 the insulator stack regions of the second stack regions overlap the gaps;
 in each of the second stack regions, at least one of the conductive layers of the second stack region is electrically connected to the source contact plugs that extend through the second stack region;
 each of the source connection lines has a line shape that extends in the second direction;
 each of the bitlines has a line shape that extends in the second direction; and
 the source contact plugs include first source contact plugs that overlap the source connection lines and second source contact plugs that overlap the bitlines.

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